

Table 1: Energy data(cm^{-1}) for Ar I relative to the ground state $3p^6\ ^1\text{S}_0$.

Configuration	Designation	LS-Term	Experimental energy	Computed energy
			$E_{exp.}$ [11]	This work
4f				
	4f [3/2] ₁	³ D ₁	120188.232	119168.67
	4f [3/2] ₂	³ D ₂	120188.631	119169.15
	4 f [5/2] ₂	¹ D ₂ [†]	120230.063	119214.71
	4 f [5/2] ₃	³ D ₃ [†]	120229.750	119214.35
	4 f [7/2] ₃	¹ F ₃	120249.876	119236.94
	4 f [7/2] ₄	³ F ₄	120249.914	119236.97
	4 f [9/2] ₄	¹ G ₄	120207.175	119190.70
	4 f [9/2] ₅	³ G ₅	120207.134	119190.53
	4f ' [5/2] ₂	³ F ₂ [†]	121654.580	120654.29
	4 f ' [5/2] ₃	³ F ₃ [†]	121654.234	120653.91
	4 f ' [7/2] ₃	³ G ₃	121653.238	120653.20
	4 f ' [7/2] ₄	³ G ₄	121653.280	120653.31
† ³ D ₃ couples with ³ F ₃ and ³ F ₂ couples with ¹ D ₂				
5d				
	5d [1/2] ₀	³ P ₀	121794.0988	121202.73
	5d [1/2] ₁	³ P ₁	121932.8477	121262.12
	5d [7/2] ₄	³ F ₄	122036.0704	121326.50
	5d [7/2] ₃	³ F ₃ [†]	122160.1502	121396.52
	5d [3/2] ₂	³ P ₂ [†]	122086.9164	121356.01
	5d [3/2] ₁	³ D ₁	122514.192	121857.63
	5d [5/2] ₂	³ F ₂	122282.104	122830.32
	5d [5/2] ₃	¹ F ₃ [†]	122329.672	122876.69
	5d' [5/2] ₂	¹ D ₂	123505.4896	121496.43
	5d' [5/2] ₃	³ D ₃	123557.4227	121524.29
	5d' [3/2] ₂	³ D ₂ [†]	123372.9253	122843.56

Table 1 (*continued*)

	$5d' [3/2]_1$	1P_1	123815.56	123687.21
	$\dagger$ 3F_3 couples with 1F_3 and 3P_2 couples with 3D_2			
6p				
	$6p[1/2]_1$	3S_1	121068.736	120160.74
	$6p[5/2]_3$	3D_3	121165.367	120256.98
	$6p[5/2]_2$	3D_2	121191.805	121743.71
	$6p[3/2]_1$	3D_1	121257.161	121718.52
	$6p[3/2]_2$	3P_2	121270.617	120362.00
	$6p[1/2]_0$	3P_0	121470.238	121937.67
	$6p' [3/2]_1$	1P_1	122609.637	120352.61
	$6p' [3/2]_2$	1D_2	122635.054	120283.28
	$6p' [1/2]_1$	3P_1	122601.225	121735.39
	$6p' [1/2]_0$	1S_0	122790.547	120570.70
7s				
	$7s[3/2]_2$	3P_2	122440.0672	121505.42
	$7s[3/2]_1$	3P_1	122479.3820	121550.17
	$7s[1/2]_0$	3P_0	123873.000	122951.58
	$7s' [1/2]_1$	1P_1	123882.203	122974.85
	3P_1 couples with 1P_1			
5f				
	$5f [3/2]_1$	3D_1	122686.065	121656.17
	$5f [3/2]_2$	3D_2	122686.410	121656.58
	$5f [5/2]_2$	$^1D_2^\dagger$	122708.126	121680.19
	$5f [5/2]_3$	$^3D_3^\dagger$	122707.902	121679.90
	$5f [7/2]_3$	1F_3	122717.823	121691.30
	$5f [7/2]_4$	3F_4	122717.853	121691.33
	$5f [9/2]_4$	1G_4	122695.697	121667.53
	$5f [9/2]_5$	3G_5	122695.636	121667.38
	$5f' [5/2]_2$	$^3F_2^\dagger$	124137.436	123122.56

Table 1 (*continued*)

$5 f' [5/2]_3$	$^3F_3^\dagger$	124137.214	123122.24
$5 f' [7/2]_3$	3G_3	124135.669	123122.05
$5 f' [7/2]_4$	3G_4	124135.698	123122.14

† 3D_3 couples with 3F_3 and 3F_2 couples with 1D_2

6d

$6d [1/2]_0$	3P_0	123508.927	122728.85
$6d [1/2]_1$	3P_1	123467.9733	122767.74
$6d [7/2]_4$	3F_4	123653.177	122802.44
$6d [7/2]_3$	$^3F_3^\dagger$	123773.8614	122846.34
$6d [3/2]_2$	$^3P_2^\dagger$	123808.513	122824.96
$6d [3/2]_1$	$^3D_1^\dagger$	123997.01	123145.10
$6d [5/2]_2$	3F_2	123826.75	124281.33
$6d [5/2]_3$	$^1F_3^\dagger$	123832.419	124310.26
$6d' [5/2]_2$	1D_2	125113.376	122905.12
$6d' [5/2]_3$	3D_3	125149.937	122921.34
$6d' [3/2]_2$	$^3D_2^\dagger$	125066.432	124285.90
$6d' [3/2]_1$	$^1P_1^\dagger$	125286.26	124500.53

† 3F_3 couples with 1F_3 , 3P_2 couples with 3D_2 and 1P_1 couples with 3D_1 .

7p

$7p [1/2]_1$	3S_1	123171.920	122185.75
$7p [5/2]_3$	3D_3	123205.780	122233.32
$7p [5/2]_2$	3D_2	123220.69	123700.90
$7p [3/2]_1$	3D_1	123254.91	123213.30
$7p [3/2]_2$	3P_2	123261.525	122288.85
$7p [1/2]_0$	3P_0	123385.044	123801.55
$7p' [3/2]_1$	1P_1	124643.45	122283.93
$7p' [3/2]_2$	1D_2	124658.44	122247.64
$7p' [1/2]_1$	3P_1	124650.95	123693.59
$7p' [1/2]_0$	1S_0	124749.805	122407.01

Table 1 (*continued*)**8s**

8s[3/2] ₂	³ P ₂	123903.220	123766.12
8s[3/2] ₁	³ P ₁ [†]	123935.870	123782.22
8s[1/2] ₀	³ P ₀	125334.69	125212.51
8s[1/2] ₁	¹ P ₁ [†]	125353.23	125220.63
[†] ³ P ₁ couples with ¹ P ₁			

Table 2: The oscillator strength f_{ij} and the transition rates A_{if} (in s^{-1}) of Ar I, as obtained in the velocity $f_{if}(v)$ and $A_{if}(v)$ and length $f_{if}(l)$ and $A_{if}(l)$ forms of the electric dipole operator in Coulomb and Babushkin gauge, respectively.

Multiplet/Terms	g_i	g_k	Type	$E(cm^{-1})$	$f_{if}(v)$	$f_{if}(l)$	$A_{if}(v)$	$A_{if}(l)$
3p⁵(²P)3d–3p⁵(²P)4f								
³ P ₀ – ³ D ₁	1	3	E ₁	6728.13	0.771305	0.977509	7.76311e ⁶	9.83853e ⁶
³ P ₁ – ³ D ₁	3	3	E ₁	6599.20	0.701916	0.855671	6.79655e ⁶	8.28533e ⁶
³ D ₁ – ³ D ₁	3	3	E ₁	5008.00	0.103938	0.073025	5.79595e ⁵	4.04721e ⁵
¹ P ₁ – ³ D ₁	3	3	E ₁	3450.21	0.010639	0.003540	2.81601e ⁴	9.37110e ³
³ P ₁ – ³ D ₂	3	5	E ₁	6599.68	1.101870	1.342830	6.40251e ⁶	7.80258e ⁶
³ P ₁ – ¹ D ₂	3	5	E ₁	6645.24	0.455585	0.563487	2.68388e ⁶	3.31954e ⁶
³ P ₁ – ³ F ₂	3	5	E ₁	8084.82	0.078629	0.143886	6.85642e ⁵	1.25468e ⁶
³ D ₁ – ³ D ₂	3	5	E ₁	5008.48	0.758486	0.532080	2.53824e ⁶	1.78058e ⁶
³ D ₁ – ¹ D ₂	3	5	E ₁	5054.04	1.664630	1.190010	5.67241e ⁶	4.05510e ⁶
³ D ₁ – ³ F ₂	3	5	E ₁	6493.62	0.436427	0.515210	2.45503e ⁶	2.89821e ⁶
¹ P ₁ – ³ D ₂	3	5	E ₁	3450.69	0.687459	0.228966	1.09202e ⁶	3.63710e ⁵
¹ P ₁ – ¹ D ₂	3	5	E ₁	3496.25	0.303941	0.103952	4.95640e ⁵	1.69516e ⁶
¹ P ₁ – ³ F ₂	3	5	E ₁	4935.83	2.450650	1.671520	7.96479e ⁶	5.43253e ⁶
³ P ₂ – ³ D ₁	5	3	E ₁	6339.60	0.071725	0.080663	6.40941e ⁵	7.20813e ⁵
³ F ₂ – ³ D ₁	5	3	E ₁	5760.79	0.035582	0.033071	2.62551e ⁵	2.44023e ⁵
¹ D ₂ – ³ D ₁	5	3	E ₁	4560.25	4.1551e ⁻³	2.4200e ⁻³	1.92125e ⁴	1.11896e ⁴
³ D ₂ – ³ D ₁	5	3	E ₁	4432.90	0.014134	7.7719e ⁻³	6.17549e ⁴	3.39568e ⁴
³ P ₂ – ³ D ₂	5	5	E ₁	6340.09	0.653991	0.735430	3.50699e ⁶	3.94370e ⁶
³ P ₂ – ¹ D ₂	5	5	E ₁	6385.65	0.187009	0.213455	1.01729e ⁶	1.16115e ⁶
³ P ₂ – ³ F ₂	5	5	E ₁	7825.22	0.023155	0.039667	1.89155e ⁵	3.24041e ⁵
³ F ₂ – ³ D ₂	5	5	E ₁	5761.27	2.8823e ⁻³	2.6643e ⁻³	1.27630e ⁴	1.17978e ⁴
³ F ₂ – ¹ D ₂	5	5	E ₁	5806.83	0.661336	0.624343	2.97490e ⁶	2.80849e ⁶
³ F ₂ – ³ F ₂	5	5	E ₁	7246.41	0.018419	0.027082	1.29028e ⁵	1.89716e ⁵
¹ D ₂ – ³ D ₂	5	5	E ₁	4560.73	2.7132e ⁻⁴	1.5709e ⁻⁴	7.52875e ²	4.35904e ²
¹ D ₂ – ¹ D ₂	5	5	E ₁	4606.29	0.084836	0.050397	2.40135e ⁵	1.42654e ⁵



Table 2 (continued)

$^1D_2 - ^3F_2$	5	5	E ₁	6045.87	0.167079	0.171011	8.14725e ⁵	8.33897e ⁵
$^3D_2 - ^3D_2$	5	5	E ₁	4433.38	0.128723	0.070779	3.37519e ⁵	1.85587e ⁵
$^3D_2 - ^1D_2$	5	5	E ₁	4478.94	0.029722	0.016691	7.95452e ⁴	4.46708e ⁴
$^3D_2 - ^3F_2$	5	5	E ₁	5918.52	0.263617	0.258351	1.23189e ⁶	1.20728e ⁶
$^3P_2 - ^3D_3$	5	7	E ₁	6385.29	2.729650	3.113430	10.6051e ⁶	12.0961e ⁶
$^3P_2 - ^1F_3$	5	7	E ₁	6407.87	1.3324e ⁻⁵	1.3130e ⁻⁵	52.1344	51.3745
$^3P_2 - ^3G_3$	5	7	E ₁	7824.14	6.2191e ⁻⁷	1.2259e ⁻⁶	3.62786	7.15164
$^3P_2 - ^3F_3$	5	7	E ₁	7824.84	0.324077	0.555055	1.89079e ⁶	3.23841e ⁶
$^3F_2 - ^3D_3$	5	7	E ₁	5806.47	0.045623	0.042947	1.46575e ⁵	1.37976e ⁵
$^3F_2 - ^1F_3$	5	7	E ₁	5829.06	3.370600	3.205900	10.9131e ⁶	10.3798e ⁶
$^3F_2 - ^3G_3$	5	7	E ₁	7245.32	0.252278	0.370757	1.26194e ⁶	1.85459e ⁶
$^3F_2 - ^3F_3$	5	7	E ₁	7246.03	1.4948e ⁻³	2.1912e ⁻³	7.74907e ³	10.9632e ⁻³
$^1D_2 - ^3D_3$	5	7	E ₁	4605.93	5.4477e ⁻³	3.2265e ⁻³	11.0127e ³	6.52246e ³
$^1D_2 - ^1F_3$	5	7	E ₁	4628.52	0.331561	0.198834	6.76849e ⁵	4.05900e ⁵
$^1D_2 - ^3G_3$	5	7	E ₁	6044.78	3.78497	3.871820	13.1786e ⁶	13.4809e ⁶
$^1D_2 - ^3F_3$	5	7	E ₁	6045.49	0.014343	0.014636	4.99542e ⁴	5.09720e ⁴
$^3D_2 - ^3D_3$	5	7	E ₁	4478.58	0.436511	0.244935	8.34298e ⁵	4.68140e ⁵
$^3D_2 - ^1F_3$	5	7	E ₁	4501.17	3.5508e ⁻⁶	1.7578e ⁻⁶	6.85536	3.393640
$^3D_2 - ^3G_3$	5	7	E ₁	5917.43	1.1627e ⁻⁴	1.1739e ⁻⁴	3.87958e ²	3.91719e ²
$^3D_2 - ^3F_3$	5	7	E ₁	5918.14	3.659779	3.585630	12.2144e ⁶	11.9668e ⁶
$^3F_3 - ^3D_2$	7	5	E ₁	6115.36	3.7343e ⁻³	3.9104e ⁻³	1.86308e ⁴	1.95094e ⁴
$^3F_3 - ^1D_2$	7	5	E ₁	6160.92	0.046675	0.049601	2.36347e ⁵	2.51165e ⁵
$^3F_3 - ^3F_2$	7	5	E ₁	7600.50	4.1596e ⁻⁴	6.7232e ⁻⁴	3.20564e ³	5.18128e ³
$^1F_3 - ^3D_2$	7	5	E ₁	5654.04	0.055269	0.049423	2.35707e ⁵	2.10776e ⁵
$^1F_3 - ^1D_2$	7	5	E ₁	5699.60	0.027659	0.025134	1.19868e ⁵	1.08926e ⁵
$^1F_3 - ^3F_2$	7	5	E ₁	7139.17	5.1787e ⁻⁴	8.3713e ⁻⁴	3.99239e ³	5.69196e ³
$^3D_3 - ^3D_2$	7	5	E ₁	4465.79	3.6877e ⁻³	2.0576e ⁻³	9.81136e ³	5.47435e ³
$^3D_3 - ^1D_2$	7	5	E ₁	4511.35	5.8217e ⁻³	3.3158e ⁻³	1.58067e ⁴	9.00280e ³
$^3D_3 - ^3F_2$	7	5	E ₁	5950.93	0.012658	0.012539	5.98043e ⁴	5.92421e ⁴

Table 2 (continued)

$^3F_3 - ^3D_3$	7	7	E ₁	6160.56	0.043016	0.045715	1.55568e ⁵	1.65328e ⁵
$^3F_3 - ^1F_3$	7	7	E ₁	6183.15	0.692961	0.741714	2.52448e ⁶	2.70209e ⁶
$^3F_3 - ^3G_3$	7	7	E ₁	7599.41	1.8442e ⁻³	2.9783e ⁻³	1.01488e ⁴	1.63900e ⁴
$^3F_3 - ^3F_3$	7	7	E ₁	7600.12	5.1021e ⁻³	8.2443e ⁻³	2.80827e ⁴	4.53778e ⁴
$^1F_3 - ^3D_3$	7	7	E ₁	5699.24	0.956972	0.869486	2.96195e ⁶	2.69117e ⁶
$^1F_3 - ^1F_3$	7	7	E ₁	5721.82	0.037089	0.033959	1.15675e ⁵	1.05943e ⁵
$^1F_3 - ^3G_3$	7	7	E ₁	7138.09	5.4213e ⁻³	7.7268e ⁻³	2.63218e ⁴	3.75154e ⁴
$^1F_3 - ^3F_3$	7	7	E ₁	7138.79	0.013006	0.018541	6.31625e ⁴	9.00405e ⁴
$^3D_3 - ^3D_3$	7	7	E ₁	4511.00	0.068713	0.039118	1.33239e ⁵	7.58521e ⁴
$^3D_3 - ^1F_3$	7	7	E ₁	4533.58	0.029793	0.017141	5.83505e ⁴	3.35719e ⁴
$^3D_3 - ^3G_3$	7	7	E ₁	5949.84	0.199338	0.197394	6.72426e ⁵	6.65871e ⁵
$^3D_3 - ^3F_3$	7	7	E ₁	5950.55	0.254916	0.252487	8.60112e ⁵	8.51916e ⁵
$^3F_3 - ^1G_4$	7	9	E ₁	6136.91	4.855580	5.119670	13.5532e ⁶	14.2903e ⁶
$^3F_3 - ^3F_4$	7	9	E ₁	6183.19	0.157217	0.168311	4.45477e ⁵	4.76909e ⁵
$^3F_3 - ^3G_4$	7	9	E ₁	7599.52	0.084147	0.135934	3.60174e ⁵	5.81837e ⁵
$^1F_3 - ^1G_4$	7	9	E ₁	5675.59	0.418144	0.376995	9.98914e ⁵	9.00031e ⁵
$^1F_3 - ^3F_4$	7	9	E ₁	5721.86	4.672050	4.278690	11.3366e ⁶	10.3821e ⁶
$^1F_3 - ^3G_4$	7	9	E ₁	7138.19	0.131407	0.187294	4.96245e ⁵	7.07295e ⁵
$^3D_3 - ^1G_4$	7	9	E ₁	4487.35	0.056298	0.031754	8.40184e ⁴	4.73897e ⁴
$^3D_3 - ^3F_4$	7	9	E ₁	4533.62	0.226176	0.130056	3.44537e ⁵	1.98116e ⁵
$^3D_3 - ^3G_4$	7	9	E ₁	5949.95	5.340170	5.288220	14.0114e ⁶	1.38751e ⁶
$^3F_4 - ^3D_3$	9	7	E ₁	6333.42	0.037586	0.042173	1.43666e ⁵	1.61198e ⁵
$^3F_4 - ^1F_3$	9	7	E ₁	6356.01	0.020812	0.023519	8.01196e ⁴	9.05399e ⁴
$^3F_4 - ^3G_3$	9	7	E ₁	7772.27	6.1491e ⁻⁶	1.0390e ⁻⁵	35.3963	59.8117
$^3F_4 - ^3F_3$	9	7	E ₁	7772.98	2.5838e ⁻⁵	4.3668e ⁻⁵	1.48758e ²	2.51411e ²
$^3F_4 - ^1G_4$	9	9	E ₁	6309.77	0.148049	0.164880	4.36852e ⁵	4.86513e ⁵
$^3F_4 - ^3F_4$	9	9	E ₁	6356.05	0.727333	0.821931	2.17775e ⁶	2.46099e ⁶
$^3F_4 - ^3G_4$	9	9	E ₁	7772.38	2.1588e ⁻⁴	3.6479e ⁻⁴	9.66550e ²	1.63328e ³

Table 2 (continued)

$^3F_4 - ^3G_5$	9	11	E ₁	6309.60	6.460740	7.194710	15.5968e ⁶	17.3686e ⁶
$3p^5(^2P)4d - 3p^5(^2P)4f$								
$^3P_0 - ^3D_1$	1	3	E ₁	787.18	0.099786	0.382551	1.37479e ⁴	5.27056e ⁴
$^3P_1 - ^3D_1$	3	3	E ₁	696.47	0.105620	0.316579	1.13914e ⁴	3.41439e ⁴
$^3D_1 - ^3D_1$	3	3	E ₁	275.34	0.017120	8.0928e ⁻³	2.88585e ²	1.36412e ²
$^1P_1 - ^3D_1$	3	3	E ₁	1715.52	1.9562e ⁻⁴	3.5139e ⁻³	1.28010e ²	2.29937e ³
$^3P_1 - ^3D_2$	3	5	E ₁	696.96	0.149901	0.448182	9.71394e ³	0.44818e ⁴
$^3P_1 - ^1D_2$	3	5	E ₁	742.52	0.072519	0.248178	5.33383e ³	1.82537e ⁴
$^3P_1 - ^3F_2$	3	5	E ₁	2182.10	1.9378e-3	0.057069	1.23096e ³	3.62513e ⁴
$^3D_1 - ^3D_2$	3	5	E ₁	274.85	0.281782	0.130470	4.73293e ³	2.19143e ³
$^3D_1 - ^1D_2$	3	5	E ₁	229.29	0.543089	0.176060	6.34855e ³	2.05810e ³
$^3D_1 - ^3F_2$	3	5	E ₁	1210.28	0.024864	0.225288	4.85870e ³	4.40236e ⁴
$^1P_1 - ^3D_2$	3	5	E ₁	1715.03	0.013452	0.242870	8.79791e ³	1.58853e ⁵
$^1P_1 - ^1D_2$	3	5	E ₁	1669.48	6.7067e ⁻³	0.115047	4.15615e ³	7.12946e ⁴
$^1P_1 - ^3F_2$	3	5	E ₁	229.90	0.875554	0.286312	1.02890e ⁴	3.36457e ³
$^3P_2 - ^3D_1$	5	3	E ₁	536.64	0.014026	0.024876	8.98145e ²	1.59287e ³
$^1D_2 - ^3D_1$	5	3	E ₁	262.22	0.012661	5.4025e ⁻³	1.93562e ²	82.5924
$^3F_2 - ^3D_1$	5	3	E ₁	1001.65	6.5254e ⁻⁵	4.0623e ⁻⁴	8.73401	54.3722
$^3D_2 - ^3D_1$	5	3	E ₁	1043.79	3.2102e ⁻⁴	2.1538e ⁻³	46.6590	3.13056e ²
$^3P_2 - ^3D_2$	5	5	E ₁	537.13	0.128152	0.227240	4.93232e ³	8.74604e ³
$^3P_2 - ^1D_2$	5	5	E ₁	582.68	0.033421	0.070098	1.51378e ³	3.17503e ³
$^3P_2 - ^3F_2$	5	5	E ₁	2022.26	2.3409e ⁻⁴	0.010783	2.34096e ²	5.88307e ³
$^1D_2 - ^3D_2$	5	5	E ₁	262.70	1.0456e ⁻³	4.2627e ⁻⁴	9.62705	3.93465
$^1D_2 - ^1D_2$	5	5	E ₁	308.26	0.203026	0.119404	2.57374e ³	1.51368e ³
$^1D_2 - ^3F_2$	5	5	E ₁	1747.84	3.6008e ⁻⁴	6.8144e ⁻³	1.46752e ²	2.77718e ³
$^3F_2 - ^3D_2$	5	5	E ₁	1001.16	3.0103e ⁻⁶	1.7593e ⁻⁵	0.402524	2.35275
$^3F_2 - ^1D_2$	5	5	E ₁	955.60	1.5481e ⁻³	8.7507e ⁻³	1.88601e ²	1.06605e ³
$^3F_2 - ^3F_2$	5	5	E ₁	483.97	0.033963	0.049299	1.06128e ³	1.54048e ³
$^3D_2 - ^3D_2$	5	5	E ₁	1043.30	2.9400e ⁻³	0.019669	4.26924e ²	2.85612e ³

Table 2 (continued)

$^3D_2 - ^1D_2$	5	5	E ₁	997.75	6.0323e ⁻⁴	3.7130e ⁻³	80.1127	4.93109e ²
$^3D_2 - ^3F_2$	5	5	E ₁	441.83	0.058564	0.070305	1.52526e ³	1.83094e ³
$^3P_2 - ^3D_3$	5	7	E ₁	582.33	0.491975	1.025210	1.58973e ⁴	3.31277e ⁴
$^3P_2 - ^1F_3$	5	7	E ₁	604.91	8.9946e ⁻⁶	9.0631e ⁻⁶	0.313624	0.316013
$^3P_2 - ^3G_3$	5	7	E ₁	2021.18	3.5586e ⁻⁹	4.7134e ⁻⁷	1.38528e ⁻³	0.183480
$^3P_2 - ^3F_3$	5	7	E ₁	2021.88	6.0075e ⁻³	0.150800	2.34019e ³	5.87461e ⁴
$^1D_2 - ^3D_3$	5	7	E ₁	307.91	0.014183	8.1225e ⁻³	1.28129e ²	73.3785
$^1D_2 - ^1F_3$	5	7	E ₁	330.49	0.958322	0.646798	9.97396e ³	6.73171e ³
$^1D_2 - ^3G_3$	5	7	E ₁	1746.75	3.02429e ⁻³	0.057979	8.79289e ²	1.65955e ⁴
$^1D_2 - ^3F_3$	5	7	E ₁	1747.46	3.0133e ⁻⁵	5.5574e ⁻⁴	8.76810	1.61709e ²
$^3F_2 - ^3D_3$	5	7	E ₁	955.96	9.1508e ⁻⁵	5.0431e ⁻⁴	11.1562	61.4832
$^3F_2 - ^1F_3$	5	7	E ₁	933.38	4.6224e ⁻³	0.024881	5.37231e ²	2.89182e ³
$^3F_2 - ^3G_3$	5	7	E ₁	482.89	0.763722	1.101440	1.69696e ⁴	2.44734e ⁴
$^3F_2 - ^3F_3$	5	7	E ₁	483.59	3.5336e ⁻³	4.9965e ⁻³	78.7454	1.11346e ²
$^3D_2 - ^3D_3$	5	7	E ₁	998.10	9.0767e ⁻³	0.055573	1.20630e ³	7.38565e ³
$^3D_2 - ^1F_3$	5	7	E ₁	975.52	5.7542e ⁻³	2.0021e ⁻⁶	7.30524e ⁻²	0.254181
$^3D_2 - ^3G_3$	5	7	E ₁	440.75	7.72502e ⁻⁵	1.0747e ⁻⁴	1.42995	1.98939
$^3D_2 - ^3F_3$	5	7	E ₁	441.45	0.807998	0.967021	1.50044e ⁴	1.79575e ⁴
$^3F_3 - ^3D_2$	7	5	E ₁	446.73	1.0210e ⁻³	1.2608e ⁻³	27.1831	33.5678
$^3F_3 - ^1D_2$	7	5	E ₁	492.29	9.4829e ⁻³	1.4219e ⁻³	3.06588e ²	4.59712e ²
$^3F_3 - ^3F_2$	7	5	E ₁	1931.87	1.4195e ⁻⁵	3.2623e ⁻⁴	7.06786	1.62424e ²
$^3D_3 - ^3D_2$	7	5	E ₁	209.30	0.023216	6.2440e ⁻³	1.35671e ²	36.4897
$^3D_3 - ^1D_2$	7	5	E ₁	254.86	9.2991e ⁻³	9.2765e ⁻⁶	80.5753	32.1308
$^3D_3 - ^3F_2$	7	5	E ₁	1694.43	9.27657e ⁻⁶	1.6352e ⁻⁴	3.55311	62.6318
$^1F_3 - ^3D_2$	7	5	E ₁	1076.14	4.5435e ⁻⁵	3.2377e ⁻⁴	5.01391	35.7292
$^1F_3 - ^1D_2$	7	5	E ₁	1030.58	1.1396e ⁻⁴	7.4650e ⁻⁴	11.5341	75.5513
$^1F_3 - ^3F_2$	7	5	E ₁	409.00	2.9792e ⁻³	3.0596e ⁻³	66.4841	68.27778
$^3F_3 - ^3D_3$	7	7	E ₁	491.93	1.1451e ⁻²	1.7148e ⁻²	2.64059e ²	3.95432e ²
$^3F_3 - ^1F_3$	7	7	E₁	514.52	0.129659	0.212361	3.27073e⁻³	5.35691e⁻³

Table 2 (continued)

$^3F_3 - ^3G_3$	7	7	E ₁	1930.78	$3.0817e^{-5}$	$7.0189e^{-4}$	10.9474	$2.49336e^2$
$^3F_3 - ^3F_3$	7	7	E ₁	1931.49	$1.4518e^{-4}$	$3.3289e^{-3}$	51.6114	$1.18340e^3$
$^3D_3 - ^3D_3$	7	7	E ₁	254.50	0.334987	0.133216	$2.06749e^3$	$8.22186e^2$
$^3D_3 - ^1F_3$	7	7	E ₁	277.08	0.010083	$4.7525e^{-3}$	73.7697	34.7688
$^3D_3 - ^3G_3$	7	7	E ₁	1693.35	$4.5582e^{-5}$	$8.0248e^{-4}$	12.4548	$2.19267e^2$
$^3D_3 - ^3F_3$	7	7	E ₁	1694.05	$2.3054e^{-4}$	$4.0623e^{-3}$	63.0465	$1.11089e^3$
$^1F_3 - ^3D_3$	7	7	E ₁	1030.94	$1.0249e^{-3}$	$6.6996e^{-3}$	$1.03840e^2$	$6.78523e^2$
$^1F_3 - ^1F_3$	7	7	E ₁	1008.36	$6.2245e^{-4}$	$3.9142e^{-3}$	60.3087	$3.79246e^2$
$^1F_3 - ^3G_3$	7	7	E ₁	407.91	0.046003	0.046996	$7.29389e^2$	$7.45140e^2$
$^1F_3 - ^3F_3$	7	7	E ₁	408.62	0.059915	0.061422	$9.53267e^2$	$9.77231e^2$
$^3F_3 - ^1G_4$	7	9	E ₁	468.28	0.954264	1.294650	$1.55090e^4$	$2.10410e^4$
$^3F_3 - ^1F_3$	7	7	E ₁	514.55	0.041296	0.067654	$8.10354e^2$	$1.32757e^3$
$^3F_3 - ^3G_4$	7	9	E ₁	1930.89	4.672050	4.278690	$5.62717e^2$	$1.28769e^4$
$^3D_3 - ^1G_4$	7	9	E ₁	230.85	0.182218	0.059623	$7.19688e^2$	$2.35487e^2$
$^3D_3 - ^1F_3$	7	7	E₁	277.12	$1.49692e^{-5}$	$0.70582e^{-3}$	$8.51990e^3$	$4.01726e^3$
$^1P_1 - ^1G_4$	3	9	E ₁	1693.45	$8.7951e^{-4}$	0.015486	$1.86934e^2$	$3.29152e^3$
$^1F_3 - ^1G_4$	7	9	E ₁	1054.59	$2.0990e^{-3}$	0.014477	$2.22453e^2$	$1.53429e^3$
$^1F_3 - ^3F_4$	7	9	E ₁	1008.32	$2.0999e^{-3}$	0.013138	$2.03448e^2$	$1.27290e^3$
$^1F_3 - ^3G_4$	7	9	E ₁	408.02	1.233920	1.261240	$1.52243e^4$	$1.55615e^4$
$^3F_4 - ^3D_3$	9	7	E ₁	606.82	$5.9858e^{-3}$	0.013533	$2.10031e^2$	$4.74847e^2$
$^3F_4 - ^1F_3$	9	7	E₁	629.40	$3.2071e^{-3}$	$7.8002e^{-3}$	$1.21063e^2$	$2.94445e^2$
$^3F_4 - ^3G_3$	9	7	E ₁	2045.67	$1.4975e^{-6}$	$3.8503e^{-5}$	0.142160	3.65243
$^3F_4 - ^3F_3$	9	7	E ₁	2046.37	$1.4975e^{-6}$	$3.8503e^{-5}$	0.597566	1.53642
$^3F_4 - ^1G_4$	9	9	E ₁	583.17	0.024443	0.051036	$6.16094e^2$	$1.28638e^3$
$^3F_4 - ^1F_3$	9	7	E ₁	629.44	0.112065	0.272607	$3.29063e^3$	$8.00469e^3$
$^3F_4 - ^3G_4$	9	9	E ₁	2045.77	$1.2514e^{-5}$	$3.2125e^{-4}$	3.88179	99.7482
$^3F_4 - ^3G_5$	9	11	E ₁	582.99	1.06690	2.226420	$2.19886e^4$	$4.5886e4$

 $3p^6 - 3p^5(^2P) 5d$

$^1S_0 - ^3P_1$	1	3	E ₁	121262.12	$1.0364e^{-3}$	$1.1706e^{-3}$	$3.38863e^6$	$3.82736e^6$
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Table 2 (continued)

$^1S_0-^3D_1$	1	3	E ₁	121857.63	0.053039	0.060041	1.75117e ⁸	1.98235e ⁸
$^1S_0-^1P_1$	1	3	E ₁	123687.21	0.068294	0.077339	2.30526e ⁸	2.61059e ⁸
$3p^5(^2P)5d-3p^5(^2P)4f$								
$^3P_0-^3D_1$	1	3	E ₁	2034.06	0.050392	0.041429	1.39070e ⁵	1.14335e ⁵
$^3P_1-^3D_1$	3	3	E ₁	2093.45	0.041429	0.046981	4.57802e ⁴	3.98715e ⁴
$^3D_1-^3D_1$	3	3	E ₁	2688.96	1.4503e ⁻³	2.0820e ⁻³	2.33162e ³	3.34714e ³
$^1P_1-^3D_1$	3	3	E ₁	4518.54	6.8801e ⁻⁵	2.2385e ⁻⁴	2.50252e ²	8.14214e ²
$^3P_1-^3D_2$	3	5	E ₁	2092.97	0.061962	0.053958	6.03498e ⁴	5.25540e ⁴
$^3P_1-^1D_2$	3	5	E ₁	2047.41	0.037954	0.031606	3.53743e ⁴	2.94581e ⁴
$^3P_1-^3F_2$	3	5	E ₁	607.83	3.2415e ⁻³	2.3798e ⁻⁴	2.66280e ²	19.5498
$^3D_1-^3D_2$	3	5	E ₁	2688.47	0.046050	0.066191	7.40058e ⁴	1.06373e ⁵
$^3D_1-^1D_2$	3	5	E ₁	2642.92	0.061725	0.085699	9.58629e ⁴	1.33096e ⁵
$^3D_1-^3F_2$	3	5	E ₁	1203.34	0.014403	4.1445e ⁻³	4.63731e ³	1.33435e ³
$^1P_1-^3D_2$	3	5	E ₁	4518.06	3.1749e ⁻³	0.010325	1.15454e ⁴	3.75469e ⁴
$^1P_1-^1D_2$	3	5	E ₁	4472.51	1.5725e ⁻³	4.9983e ⁻³	5.59041e ³	1.77690e ⁴
$^1P_1-^3F_2$	3	5	E ₁	3032.92	0.112601	0.146523	1.63950e ⁵	2.13342e ⁵
$^3P_2-^3D_1$	5	3	E ₁	2187.35	4.6105e-3	4.3849e-3	2.94281e ³	2.79882e ³
$^1D_2-^3D_1$	5	3	E ₁	2327.76	1.8925e-3	2.0372e ⁻³	1.36807e ³	1.47264e ³
$^3F_2-^3D_1$	5	3	E ₁	3661.67	6.4716e ⁻⁶	1.7238e ⁻⁵	11.5756	30.8335
$^3D_2-^3D_1$	5	3	E ₁	3674.91	3.8743e ⁻⁵	1.0401e ⁻⁴	69.8023	1.87388e ²
$^3P_2-^3D_2$	5	5	E ₁	2186.86	0.042132	0.040060	2.68800e ⁴	2.55583e ⁴
$^3P_2-^1D_2$	5	5	E ₁	2141.30	0.012127	0.011050	7.41812e ³	6.75944e ³
$^3P_2-^3F_2$	5	5	E ₁	701.73	4.2006e ⁻⁴	4.1127e ⁻⁵	27.5946	2.70168
$^1D_2-^3D_2$	5	5	E ₁	2327.28	1.4999e ⁻⁴	1.6204e ⁻⁴	1.08360e ²	1.17085e ²
$^1D_2-^1D_2$	5	5	E ₁	2281.72	0.036385	0.037641	2.52712e ⁴	2.61430e ⁴
$^1D_2-^3F_2$	5	5	E ₁	842.14	4.2036e ⁻⁴	5.9234e ⁻⁵	39.7713	5.60428
$^3F_2-^3D_2$	5	5	E ₁	3661.18	1.1414e ⁻⁷	3.0634e ⁻⁷	0.204119	0.547802
$^3F_2-^1D_2$	5	5	E ₁	3615.63	1.7089e ⁻⁴	4.4389e ⁻⁴	2.98029e ²	7.74145e ²
$^3F_2-^3F_2$	5	5	E ₁	2176.05	9.8770e ⁻³	9.2922e ⁻³	6.23928e ³	5.86986e ³

Table 2 (continued)

$^3D_2 - ^3D_2$	5	5	E ₁	3674.43	$3.5564e^{-4}$	$9.5467e^{-4}$	$6.40570e^2$	$1.71951e^3$
$^3D_2 - ^1D_2$	5	5	E ₁	3628.87	$5.0630e^{-5}$	$1.3248e^{-4}$	88.9462	$2.32736e^2$
$^3D_2 - ^3F_2$	5	5	E ₁	2189.29	0.015949	0.015198	$1.04198e^4$	$9.71771e^3$
$^3P_2 - ^3D_3$	5	7	E ₁	2141.66	0.017799	0.016232	$1.08914e^5$	$9.93253e^4$
$^3P_2 - ^1F_3$	5	7	E ₁	2119.08	$2.5360e^{-6}$	$2.4133e^{-6}$	1.51921	1.44572
$^3P_2 - ^3G_3$	5	7	E ₁	702.81	$1.2755e^{-8}$	$1.0821e^{-9}$	$8.40498e^{-4}$	$7.13075e^{-5}$
$^3P_2 - ^3F_3$	5	7	E ₁	702.10	$5.8739e^{-3}$	$5.7575e^{-4}$	$3.86285e^2$	37.8626
$^1D_2 - ^3D_3$	5	7	E ₁	2282.08	$2.4860e^{-3}$	$2.5779e^{-3}$	$1.72719e^3$	$1.79106e^3$
$^1D_2 - ^1F_3$	5	7	E ₁	2259.49	0.184805	0.187498	$1.25866e^5$	$1.27700e^5$
$^1D_2 - ^3G_3$	5	7	E ₁	843.23	$1.6045e^{-3}$	$2.2670e^{-4}$	$1.52200e^2$	21.5041
$^1D_2 - ^3F_3$	5	7	E ₁	842.52	$3.3681e^{-5}$	$4.7606e^{-6}$	3.18954	0.450819
$^3F_2 - ^3D_3$	5	7	E ₁	3615.98	$8.8138e^{-6}$	$2.2952e^{-5}$	15.3741	40.0370
$^3F_2 - ^1F_3$	5	7	E ₁	3593.40	$2.8797e^{-4}$	$7.3898e^{-4}$	$4.96064e^2$	$1.27297e^3$
$^3F_2 - ^3G_3$	5	7	E ₁	2177.13	0.223495	0.210507	$1.41322e^5$	$1.33110e^5$
$^3F_2 - ^3F_3$	5	7	E ₁	2176.43	$1.2860e^{-3}$	$1.2127e^{-3}$	$8.12690e^2$	$7.66346e^2$
$^3D_2 - ^3D_3$	5	7	E ₁	3629.22	$7.9024e^{-4}$	$2.0694e^{-3}$	$1.38855e^3$	$3.63620e^3$
$^3D_2 - ^1F_3$	5	7	E ₁	3606.64	$9.9495e^{-8}$	$2.6575e^{-7}$	0.172647	0.461159
$^3D_2 - ^3G_3$	5	7	E ₁	2190.38	$7.2201e^{-5}$	$6.8360e^{-5}$	46.2120	43.7537
$^3D_2 - ^3F_3$	5	7	E ₁	2189.67	0.216659	0.206550	$1.38582e^5$	$1.32115e^5$
$^3F_3 - ^3D_2$	7	5	E ₁	2227.37	$2.9025e^{-4}$	$2.8614e^{-4}$	$1.37217e^2$	$1.35272e^2$
$^3F_3 - ^1D_2$	7	5	E ₁	2181.81	$2.8591e^{-3}$	$2.7045e^{-3}$	$1.29692e^3$	$1.22678e^3$
$^3F_3 - ^3F_2$	7	5	E ₁	742.24	$2.7199e^{-5}$	$2.9874e^{-6}$	1.42785	0.156356
$^3D_3 - ^3D_2$	7	5	E ₁	2355.14	$2.6776e^{-3}$	$2.9530e^{-3}$	$1.41522e^3$	$1.56083e^3$
$^3D_3 - ^1D_2$	7	5	E ₁	2309.58	$1.3122e^{-3}$	$1.3917e^{-3}$	$6.66986e^2$	$7.07401e^2$
$^3D_3 - ^3F_2$	7	5	E ₁	870.00	$1.0316e^{-5}$	$1.5525e^{-6}$	0.744065	0.111979
$^1F_3 - ^3D_2$	7	5	E ₁	3707.55	$4.5463e^{-6}$	$1.2423e^{-5}$	5.95502	16.2727
$^1F_3 - ^1D_2$	7	5	E ₁	3661.99	$1.4756e^{-5}$	$3.9331e^{-5}$	18.8564	50.2596
$^1F_3 - ^3F_2$	7	5	E ₁	2222.41	$7.2417e^{-4}$	$7.1117e^{-4}$	$3.40829e^2$	$3.34712e^2$
$^3F_3 - ^3D_3$	7	7	E ₁	2182.17	$3.7286e^{-3}$	$3.5281e^{-3}$	$1.69189e^3$	$1.60093e^3$

Table 2 (continued)

$^3F_3 - ^1F_3$	7	7	E ₁	2159.59	0.040701	0.037719	1.80882e ⁴	1.67630e ⁴
$^3F_3 - ^3G_3$	7	7	E ₁	743.32	1.1372e ⁻⁵	1.2508e ⁻⁶	0.598734	0.065854
$^3F_3 - ^3F_3$	7	7	E ₁	742.62	2.0979e ⁻⁴	2.3001e ⁻⁵	11.0247	1.20874
$^3D_3 - ^3D_3$	7	7	E ₁	2309.94	0.047988	0.059141	2.43996e ⁴	2.58871e ⁴
$^3D_3 - ^1F_3$	7	7	E ₁	2287.36	1.4741e ⁻³	1.5334e ⁻³	7.34940e ²	7.64526e ²
$^3D_3 - ^3G_3$	7	7	E ₁	871.09	2.0263e ⁻⁵	3.0572e ⁻⁶	1.46519	0.221056
$^3D_3 - ^3F_3$	7	7	E ₁	870.38	2.8606e ⁻⁴	4.3090e ⁻⁵	20.6502	3.11059
$^1F_3 - ^3D_3$	7	7	E ₁	3662.35	1.2593e ⁻⁴	3.3582e ⁻⁴	1.60961e ²	4.29213e ²
$^1F_3 - ^1F_3$	7	7	E ₁	3639.76	6.3480e ⁻⁵	1.6710e ⁻⁴	80.1370	2.10949e ²
$^1F_3 - ^3G_3$	7	7	E ₁	2223.50	0.011028	0.010841	5.19572e ³	5.10742e ³
$^1F_3 - ^3F_3$	7	7	E ₁	2222.79	0.014514	0.014259	6.83364e ³	6.71349e ³
$^3F_3 - ^1G_4$	7	9	E ₁	2205.82	0.263854	0.255106	1.22334e ⁵	1.18278e ⁵
$^3F_3 - ^3F_4$	7	9	E ₁	2159.55	0.014224	0.013181	6.32111e ³	5.85794e ³
$^3F_3 - ^3G_4$	7	9	E ₁	743.22	2.2569e ⁻³	2.4790e ⁻⁴	1.18795e ²	13.0483
$^3D_3 - ^1G_4$	7	9	E ₁	2333.59	0.024643	0.026683	1.27878e ⁴	1.38463e ⁴
$^3D_3 - ^3F_4$	7	9	E ₁	2287.32	0.023528	0.024476	1.17296e ⁵	1.22022e ⁵
$^3D_3 - ^3G_4$	3	9	E ₁	870.98	2.1264e ⁻⁴	3.2075e ⁻⁵	15.3713	2.31863
$^1F_3 - ^1G_4$	7	9	E ₁	3686.00	3.0432e ⁻⁴	8.2147e ⁻⁴	3.93996e ²	1.06353e ³
$^1F_3 - ^3F_4$	7	9	E ₁	3639.73	9.9057e ⁻⁵	2.6084e ⁻⁴	1.25045e ²	3.29281e ²
$^1F_3 - ^3G_4$	7	9	E ₁	2223.39	0.296413	0.291360	1.39629e ⁵	1.37248e ⁵
$^3F_4 - ^3D_3$	9	7	E ₁	2112.15	2.2334e ⁻³	1.9811e ⁻³	7.38463e ²	6.55047e ²
$^3F_4 - ^1F_3$	9	7	E ₁	2089.56	1.2546e ⁻³	1.0891e ⁻³	4.05991e ²	3.52458e ²
$^3F_4 - ^3G_3$	9	7	E ₁	673.30	1.4067e ⁻⁶	1.2679e ⁻⁷	0.0472632	4.26007e ⁻³
$^3F_4 - ^3F_3$	9	7	E ₁	672.59	5.9174e ⁻⁶	5.3226e ⁻⁷	0.198396	0.0178456
$^3F_4 - ^1G_4$	9	9	E ₁	2135.80	8.6678e ⁻³	7.8616e ⁻³	2.93044e ³	2.65785e ³
$^3F_4 - ^3F_4$	9	7	E ₁	2089.52	0.043843	0.038062	1.41872e ⁴	1.23166e ⁴
$^3F_4 - ^3G_4$	9	9	E ₁	673.19	4.9392e ⁻⁵	4.4508e ⁻⁶	1.65898	0.149491
$^3F_4 - ^3G_5$	9	11	E ₁	2135.97	0.378195	0.343090	1.27881e ⁵	1.16011e ⁵

$3p^5(^2P) 5d - 3p^5(^2P) 6p$

Table 2 (continued)

$^3P_0 - ^3S_1$	1	3	E ₁	1041.99	0.889186	0.716909	6.43963e ⁵	5.19197e ⁵
$^3P_0 - ^1P_1$	1	3	E ₁	850.12	0.201893	0.107098	9.73256e ⁴	5.16280e ⁴
$^3P_0 - ^3D_1$	1	3	E ₁	515.79	2.3577e ⁻⁵	4.4495e ⁻⁶	1.39466	0.263196
$^3P_0 - ^3P_1$	1	3	E ₁	532.66	0.014442	3.3042e ⁻³	9.11111e ²	1.91941e ²
$^3P_1 - ^1S_0$	3	1	E ₁	691.43	0.093519	0.033534	9.94058e ³	3.56456e ³
$^3P_1 - ^3P_0$	3	1	E ₁	75.54	0.015317	5.1524e ⁻³	4.66275e ³	1.56841e ³
$^3D_1 - ^1S_0$	3	1	E ₁	1286.93	1.35220	1.76699	4.97934e ⁵	6.17537e ⁵
$^3D_1 - ^3P_0$	3	1	E ₁	80.04	0.287079	1.3450e ⁻³	1.22673e ³	5.74754
$^1P_1 - ^1S_0$	3	1	E ₁	3116.51	0.010526	0.055666	1.63448e ⁴	8.64333e ⁴
$^1P_1 - ^3P_0$	3	1	E ₁	749.54	1.42007	1.71021	5.13789e ⁵	6.18764e ⁵
$^3P_1 - ^3S_1$	3	3	E ₁	1101.38	2.05341	1.85195	5.53822e ⁵	4.99488e ⁵
$^3P_1 - ^1P_1$	3	3	E ₁	909.51	0.188206	0.113302	3.46156e ⁴	2.08391e ⁴
$^3P_1 - ^3D_1$	3	3	E ₁	456.40	0.011591	1.7751e ⁻³	5.36852e ²	82.2139
$^3P_1 - ^3P_1$	3	3	E ₁	473.27	6.0691e ⁻³	1.0046e ⁻³	3.02254e ²	50.0319
$^3D_1 - ^3S_1$	3	3	E ₁	1696.89	0.017902	0.040027	1.14612e ⁴	2.56262e ⁴
$^3D_1 - ^1P_1$	3	3	E ₁	1505.02	0.891352	1.50380	4.48905e ⁵	7.75349e ⁵
$^3D_1 - ^3D_1$	3	3	E ₁	139.11	0.124130	1.7676e ⁻³	5.34066e ²	7.60506
$^3D_1 - ^3P_1$	3	3	E ₁	122.23	0.254515	2.8274e ⁻³	8.45512e ²	9.39304
$^1P_1 - ^3S_1$	3	3	E ₁	3526.47	3.0653e ⁻³	0.021561	6.35081e ³	4.46709e ⁴
$^1P_1 - ^1P_1$	3	3	E ₁	3334.60	0.013063	0.080106	2.37689e ⁴	1.45758e ⁵
$^1P_1 - ^3D_1$	3	3	E ₁	2268.69	0.402181	0.663551	1.99801e ⁵	3.29649e ⁵
$^1P_1 - ^3P_1$	3	3	E ₁	1951.82	0.497754	0.811390	2.41730e ⁵	3.94046e ⁵
$^3P_1 - ^1D_2$	3	5	E ₁	978.84	0.156524	0.110364	3.33447e ⁴	2.35111e ⁴
$^3P_1 - ^3P_2$	3	5	E ₁	900.12	0.712097	0.430873	1.28280e ⁵	7.76195e ⁴
$^3P_1 - ^3D_2$	3	5	E ₁	481.59	6.3062e ⁻³	1.0908e ⁻³	1.96118e ²	33.7519
$^3D_1 - ^1D_2$	3	5	E ₁	1574.35	0.074210	0.136235	4.08966e ⁴	7.50778e ⁴
$^3D_1 - ^3P_2$	3	5	E ₁	1495.63	0.638110	0.961630	63.8110	96.1630
$^3D_1 - ^3F_2$	3	5	E ₁	113.91	0.029278	2.8481e ⁻³	84.4743	0.821753
$^1P_1 - ^1D_2$	3	5	E ₁	3403.93	3.3033e ⁻⁴	2.1123e ⁻³	6.30540e ²	4.03203e ³

Table 2 (*continued*)

$^1P_1 - ^3P_2$	3	5	E ₁	3325.21	$2.3543e^{-3}$	0.014351	$4.25581e^3$	$2.59413e^4$
$^1P_1 - ^3D_2$	3	5	E ₁	1943.50	0.058510	0.094550	$2.80961e^4$	$4.54021e^4$
$^3P_2 - ^3S_1$	5	3	E ₁	1195.27	1.91181	2.039740	$3.64378e^5$	$3.88759e^5$
$^3P_2 - ^1P_1$	5	3	E ₁	1003.41	0.318472	0.242641	$4.27758e^4$	$3.25906e^4$
$^3P_2 - ^3D_1$	5	3	E ₁	362.51	$4.2231e^{-3}$	$4.2392e^{-4}$	$1.23391e^2$	12.3862
$^3P_2 - ^3P_1$	5	3	E ₁	379.38	0.039410	$4.2608e^{-3}$	$1.26118e^3$	$1.36352e^2$
$^1D_2 - ^3S_1$	5	3	E ₁	1335.69	$1.1025e^{-3}$	$1.0734e^{-3}$	$2.62407e^2$	$2.55472e^2$
$^1D_2 - ^1P_1$	5	3	E ₁	1143.82	3.03174	2.95063	$5.29156e^5$	$5.14999e^5$
$^1D_2 - ^3D_1$	5	3	E ₁	222.09	0.040584	$1.4762e^{-3}$	$4.45069e^2$	16.1891
$^1D_2 - ^3P_1$	5	3	E ₁	238.96	$5.7402e^{-4}$	$2.7633e^{-5}$	7.28792	0.350837
$^3F_2 - ^3S_1$	5	3	E ₁	2669.58	$1.0335e^{-6}$	$1.5645e^{-6}$	0.982619	1.48751
$^3F_2 - ^1P_1$	5	3	E ₁	2477.71	$1.2838e^{-3}$	$5.8932e^{-3}$	$1.05143e^3$	$4.82646e^3$
$^3F_2 - ^3D_1$	5	3	E ₁	1111.80	4.406170	4.033430	$7.26589e^5$	$6.65122e^5$
$^3F_2 - ^3P_1$	5	3	E ₁	1094.93	0.083285	0.076109	$1.33203e^4$	$1.21727e^4$
$^3D_2 - ^3S_1$	5	3	E ₁	2682.82	$7.8209e^{-5}$	$4.3364e^{-4}$	75.0961	$4.16377e^2$
$^3D_2 - ^1P_1$	5	3	E ₁	2490.96	$2.2090e^{-3}$	0.010387	$1.82855e^3$	$8.59815e^3$
$^3D_2 - ^3D_1$	5	3	E ₁	1125.04	$6.6120e^{-7}$	$1.1473e^{-5}$	0.111647	1.93737
$^3D_2 - ^3P_1$	5	3	E ₁	1108.17	4.16531	3.83010	$6.82389e^5$	$6.27474e^5$
$^3P_2 - ^1D_2$	5	5	E ₁	1072.74	0.027577	0.022800	$4.23369e^3$	$3.50022e^3$
$^3P_2 - ^3P_2$	5	5	E ₁	994.01	2.59327	1.91695	$3.41825e^5$	$2.52678e^5$
$^3P_2 - ^3D_2$	5	5	E ₁	387.70	0.055824	$6.2764e^{-3}$	$1.11941e^3$	$1.25856e^2$
$^1D_2 - ^1D_2$	5	5	E ₁	1213.15	1.47773	1.60929	$2.90133e^5$	$3.15964e^5$
$^1D_2 - ^3P_2$	5	5	E ₁	1134.43	0.125555	0.119353	$2.15557e^4$	$2.04909e^4$
$^1D_2 - ^3D_2$	5	5	E ₁	247.28	$7.5883e^{-4}$	$3.5803e^{-5}$	6.19013	0.292065
$^3F_2 - ^1D_2$	5	5	E ₁	2547.04	$2.41663e^{-3}$	0.011573	$2.09148e^3$	$1.00163e^4$
$^3F_2 - ^3P_2$	5	5	E ₁	2468.32	$7.5190e^{-5}$	$3.3433e^{-4}$	61.1136	$2.71739e^2$
$^3F_2 - ^3D_2$	5	5	E ₁	1086.61	0.485611	0.428887	$7.64904e^4$	$6.75557e^4$
$^3D_2 - ^1D_2$	5	5	E ₁	2560.28	$6.2470e^{-4}$	$3.0035e^{-3}$	$5.46286e^2$	$2.62656e^3$
$^3D_2 - ^3P_2$	5	5	E ₁	2481.56	0.018081	0.083314	$1.48546e^4$	$6.84455e^4$

Table 2 (continued)

$^3D_2-^3D_2$	5	5	E ₁	1099.85	0.752005	0.681365	1.21356e ⁵	1.09956e ⁵
$^3P_2-^3D_3$	5	7	E ₁	1099.04	0.229301	0.207400	3.69488e ⁴	3.34198e ⁴
$^1D_2-^3D_3$	5	7	E ₁	1239.45	0.089160	0.102438	1.82728e ⁴	2.0994e ⁴
$^3F_2-^3D_3$	5	7	E ₁	2573.34	2.3096e ⁻⁴	1.1438e ⁻³	2.04035e ²	1.01053e ³
$^3D_2-^3D_3$	5	7	E ₁	2586.58	1.3755e ⁻³	6.8913e ⁻³	1.22771e ³	6.15080e ³
$^3F_3-^1D_2$	7	5	E ₁	1113.25	5.97128	5.47841	7.05170e ⁵	6.46965e ⁵
$^3F_3-^3P_2$	7	5	E ₁	1034.52	0.042923	0.033143	4.37741e ³	3.38005e ³
$^3F_3-^3D_2$	7	5	E ₁	347.19	0.023094	2.1056e ⁻³	3.71367e ²	33.8594
$^3D_3-^1D_2$	7	5	E ₁	1241.01	0.377613	0.435542	5.54172e ⁴	6.39187e ⁴
$^3D_3-^3P_2$	7	5	E ₁	1162.29	4.911470	4.971100	6.32245e ⁵	6.39922e ⁵
$^3D_3-^3P_2$	7	5	E ₁	219.42	0.014478	5.2218e ⁻⁴	92.9941	3.35390
$^1F_3-^1D_2$	7	5	E ₁	2593.41	1.2118e ⁻³	5.9254e ⁻³	7.76647e ²	3.79759e ³
$^1F_3-^3P_2$	7	5	E ₁	2514.69	3.8654e ⁻⁴	1.8062e ⁻³	2.32922e ²	1.08841e ³
$^1F_3-^3D_2$	7	5	E ₁	1132.97	6.78083	6.52099	8.29404e ⁵	7.97622e ⁵
$^3F_3-^3D_3$	7	7	E ₁	1139.55	0.913488	0.887604	1.13034e ⁵	1.09831e ⁵
$^3D_3-^3D_3$	7	7	E ₁	1267.31	1.19595	1.43907	1.83031e ⁵	2.20239e ⁵
$^1F_3-^3D_3$	7	7	E ₁	2619.71	3.6322e ⁻³	0.018662	2.37531e ³	1.22043e ⁴
$^3F_4-^3D_3$	9	7	E ₁	1069.52	9.25165	7.92863	7.84330e ⁵	6.72167e ⁵
3p⁶-3p⁵(²P)7s								
$^1S_0-^3P_1$	1	3	E ₁	121550.17	5.6279e ⁻³	6.1725e ⁻³	1.84877e ⁷	2.02767e ⁷
$^1S_0-^1P_1$	1	3	E ₁	122974.85	3.0371e ⁻³	3.3239e ⁻³	1.02123e ⁷	1.11765e ⁷
3p⁵(²P)7s-3p⁵(²P) 6p								
$^3P_0-^3S_1$	1	3	E ₁	2790.84	1.6436e ⁻³	8.7649e ⁻³	8.53941e ³	4.55599e ⁴
$^3P_0-^1P_1$	1	3	E ₁	2598.97	4.3029e ⁻⁴	1.9683e ⁻³	1.93869e ³	8.86825e ³
$^3P_0-^3D_1$	1	3	E ₁	1233.06	0.744289	0.765893	7.54838e ⁵	7.76748e ⁵
$^3P_0-^3P_1$	1	3	E ₁	1216.19	0.612591	0.619702	6.11403e ⁵	6.11403e ⁵
$^1P_1-^1S_0$	3	1	E ₁	979.48	0.563324	0.368848	1.20163e ⁵	7.86789e ⁴
$^1P_1-^3P_0$	3	1	E ₁	387.49	0.010741	1.1039e ⁻³	1.07583e ³	1.10566e ²
$^3P_1-^1S_0$	3	1	E ₁	2404.15	1.7346e ⁻³	6.7482e ⁻³	2.22926e ³	8.67230e ³

Table 2 (continued)

$^3P_1 - ^3P_0$	3	1	E ₁	1037.18	0.532660	0.386782	1.27403e ⁵	9.25114e ⁴
$^1P_1 - ^3S_1$	3	3	E ₁	1389.43	0.185178	0.245888	7.94850e ⁴	1.05544e ⁵
$^1P_1 - ^1P_1$	3	3	E ₁	1197.57	1.17719	1.14812	3.75377e ⁵	3.66108e ⁵
$^1P_1 - ^3D_1$	3	3	E ₁	168.35	3.6641e ⁻³	7.1321e ⁻⁵	23.0885	0.449410
$^1P_1 - ^3P_1$	3	3	E ₁	185.22	2.4706e ⁻³	5.8995e ⁻³	18.8454	0.449995
$^3P_1 - ^3S_1$	3	3	E ₁	2814.10	2.4888e ⁻³	0.013382	4.38234e ³	2.35629e ⁴
$^3P_1 - ^1P_1$	3	3	E ₁	2622.24	6.2956e ⁻⁸	5.2601e ⁻⁷	0.096251	0.804202
$^3P_1 - ^3D_1$	3	3	E ₁	1256.32	0.598102	0.635877	2.09894e ⁵	2.23151e ⁵
$^3P_1 - ^3P_1$	3	3	E ₁	1239.45	0.734824	0.768433	2.50995e ⁵	2.62475e ⁵
$^1P_1 - ^1D_2$	3	5	E ₁	1266.90	1.66531	1.80853	5.94290e ⁵	6.45399e ⁵
$^1P_1 - ^3P_2$	3	5	E ₁	1188.17	0.564799	0.546525	1.77286e ⁵	1.71550e ⁵
$^1P_1 - ^3D_2$	3	5	E ₁	193.54	2.8073e ⁻³	7.2871e ⁻⁵	14.0284	0.364135
$^3P_1 - ^1D_2$	3	5	E ₁	2691.57	7.1298e ⁻⁶	3.7562e ⁻⁵	11.4845	60.5041
$^3P_1 - ^3P_2$	3	5	E ₁	2612.84	1.1674e ⁻³	5.4472e ⁻³	1.77205e ³	8.26854e ³
$^3P_1 - ^3D_2$	3	5	E ₁	1231.13	2.25529	2.33303	7.60034e ⁵	7.86233e ⁵
$^3P_2 - ^3S_1$	5	3	E ₁	1344.67	1.04105	1.27518	2.51117e ⁵	3.07595e ⁵
$^3P_2 - ^1P_1$	5	3	E ₁	1152.81	0.223836	0.199221	3.96841e ⁴	3.53201e ⁴
$^3P_2 - ^3D_1$	5	3	E ₁	213.10	8.4565e ⁻⁵	2.4860e ⁻⁶	0.853885	0.025102
$^3P_2 - ^3P_1$	5	3	E ₁	229.98	0.050538	1.8105e ⁻³	5.94308e ²	21.2909
$^3P_2 - ^1D_2$	5	5	E ₁	1222.14	0.550772	0.552902	1.09745e ⁵	1.10169e ⁵
$^3P_2 - ^3P_2$	5	5	E ₁	1143.41	1.84164	1.63937	3.21206e ⁵	2.85929e ⁵
$^3P_2 - ^3D_2$	5	5	E ₁	238.30	0.010608	4.0823e ⁻⁴	80.3617	3.09254
$^3P_2 - ^3D_3$	5	7	E ₁	1248.44	3.11708	3.31583	6.48117e ⁵	6.89441e ⁵

 $3p^5(^2P)3d-3p^5(^2P)5f$

$^3P_0 - ^3D_1$	1	3	E ₁	9215.64	0.153539	0.183166	2.89928e ⁶	3.45872e ⁶
$^3P_1 - ^3D_1$	3	3	E ₁	9086.71	0.139010	0.161208	2.55199e ⁶	2.95952e ⁶
$^3D_1 - ^3D_1$	3	3	E ₁	7495.51	0.018926	0.014942	2.36419e ⁵	1.86653e ⁵
$^1P_1 - ^3D_1$	3	3	E ₁	537.71	1.6875e ⁻³	8.3492e ⁻⁴	1.32288e ⁴	6.54495e ³
$^3P_1 - ^3D_2$	3	5	E ₁	9087.11	0.219750	0.254787	2.42077e ⁶	2.80673e ⁶

Table 2 (continued)

$^3P_1 - ^1D_2$	3	5	E ₁	9110.72	0.087818	0.102539	9.73550e ⁵	1.13545e ⁶
$^3P_1 - ^3F_2$	3	5	E ₁	10553.09	0.017470	0.027329	2.59559e ⁵	4.06038e ⁵
$^3D_1 - ^3D_2$	3	5	E ₁	7495.91	0.136077	0.107319	1.02002e ⁶	8.04444e ⁵
$^3D_1 - ^1D_2$	3	5	E ₁	7519.52	0.312287	0.247975	2.35562e ⁶	1.87051e ⁶
$^3D_1 - ^3F_2$	3	5	E ₁	8961.89	0.082090	0.092611	8.79561e ⁵	9.92288e ⁵
$^1P_1 - ^3D_2$	3	5	E ₁	5938.12	0.108351	0.053633	5.09685e ⁵	2.52293e ⁵
$^1P_1 - ^1D_2$	3	5	E ₁	5961.73	0.044819	0.022366	2.12514e ⁵	1.06051e ⁵
$^1P_1 - ^3F_2$	3	5	E ₁	7404.10	0.449027	0.345774	3.28389e ⁶	2.52877e ⁶
$^3P_2 - ^3D_1$	5	3	E ₁	8827.11	0.014051	0.015373	2.43428e ⁵	2.66337e ⁵
$^3F_2 - ^3D_1$	5	3	E ₁	8248.29	6.7741e ⁻³	6.4753e ⁻³	10.2472e ⁴	9.79509e ⁴
$^1D_2 - ^3D_1$	5	3	E ₁	7047.75	7.3288e ⁻⁴	5.1145e ⁻⁴	8.09394e ³	5.64850e ³
$^3D_2 - ^3D_1$	5	3	E ₁	6920.40	2.4696e ⁻³	1.6607e ⁻⁴	2.62973e ⁴	1.76847e ⁴
$^3P_2 - ^3D_2$	5	5	E ₁	8827.51	0.128855	0.140971	1.33952e ⁶	1.46548e ⁶
$^3P_2 - ^1D_2$	5	5	E ₁	8851.12	0.035663	0.039241	3.72723e ⁵	4.10122e ⁵
$^3P_2 - ^3F_2$	5	5	E ₁	10293.50	5.1643e ⁻³	7.6829e ⁻³	7.29990e ⁴	1.08595e ⁵
$^3F_2 - ^3D_2$	5	5	E ₁	8248.70	4.6570e ⁻⁴	4.4337e ⁻⁴	4.22720e ³	4.02450e ³
$^3F_2 - ^1D_2$	5	5	E ₁	8272.31	0.127302	0.122369	1.16214e ⁶	1.11712e ⁶
$^3F_2 - ^3F_2$	5	5	E ₁	9714.68	3.1988e ⁻³	4.2410e ⁻³	4.02738e ⁴	5.33952e ⁴
$^1D_2 - ^3D_2$	5	5	E ₁	7048.16	4.0124e ⁻⁵	2.7879e ⁻⁵	2.65908e ²	1.84758e ²
$^1D_2 - ^1D_2$	5	5	E ₁	7071.77	0.014394	0.010112	9.60364e ⁴	6.74648e ⁴
$^1D_2 - ^3F_2$	5	5	E ₁	8514.14	0.032909	0.033513	3.18252e ⁵	3.24100e ⁵
$^3D_2 - ^3D_2$	5	5	E ₁	6920.81	0.022628	0.015216	1.44592e ⁵	9.72324e ⁴
$^3D_2 - ^1D_2$	5	5	E ₁	6944.42	5.6253e ⁻³	3.8103e ⁻³	3.61906e ⁴	2.45136e ⁴
$^3D_2 - ^3F_2$	5	5	E ₁	8386.79	0.050311	0.049686	4.72099e ⁵	4.66231e ⁵
$^3P_2 - ^3D_3$	5	7	E ₁	8850.83	0.531356	0.584387	3.96641e ⁶	4.36227e ⁶
$^3P_2 - ^3F_4$	5	9	E ₁	8862.23	2.8207e ⁻⁶	2.8176e ⁻⁶	21.1101	21.0871
$^3P_2 - ^3G_3$	5	7	E ₁	10292.98	3.4079e ⁻⁸	6.0441e ⁻⁸	0.344044	0.610187
$^3P_2 - ^3F_3$	5	7	E ₁	10293.17	0.072294	0.107530	7.29876e ⁵	10.8561e ⁵
$^3F_2 - ^3D_3$	5	7	E ₁	8272.02	8.8047e ⁻³	8.4471e ⁻³	5.74096e ⁴	5.50778e ⁴

Table 2 (continued)

$^3F_2 - ^1F_3$	5	7	E ₁	8283.42	0.643034	0.619708	4.20434e ⁶	4.05183e ⁶
$^3F_2 - ^3G_3$	5	7	E ₁	9714.17	0.054509	0.072252	4.90152e ⁵	6.49692e ⁵
$^3F_2 - ^3F_3$	5	7	E ₁	9714.36	2.5858e ⁻⁴	3.4211e ⁻⁴	2.32525e ³	3.07638e ³
$^1D_2 - ^3D_3$	5	7	E ₁	7071.48	9.2020e ⁻⁴	6.4513e ⁻⁴	4.38480e ³	3.07408e ³
$^1D_2 - ^1F_3$	5	7	E ₁	7082.88	0.063433	0.044695	3.03239e ⁵	2.13667e ⁵
$^1D_2 - ^3G_3$	5	7	E ₁	8513.63	0.729204	0.742408	5.03643e ⁶	5.12763e ⁶
$^1D_2 - ^3F_3$	5	7	E ₁	8513.82	2.7655e ⁻³	2.8104e ⁻³	1.91020e ⁴	1.94122e ⁴
$^3D_2 - ^3D_3$	5	7	E ₁	6944.13	0.084212	0.057011	3.86953e ⁵	2.61963e ⁵
$^3D_2 - ^1F_3$	5	7	E ₁	6955.53	7.4113e ⁻⁷	4.6387e ⁻⁷	3.41668	2.13847
$^3D_2 - ^3G_3$	5	7	E ₁	8386.28	1.6944e ⁻⁵	1.7132e ⁻⁵	1.13557e ²	1.14813e ²
$^3D_2 - ^3F_3$	5	7	E ₁	8386.47	0.698512	6.89695	4.68140e ⁶	4.62231e ⁶
$^3F_3 - ^3D_2$	7	5	E ₁	8602.79	7.5118e ⁻⁴	7.8098e ⁻⁴	7.41651e ³	7.71068e ³
$^3F_3 - ^1D_2$	7	5	E ₁	8626.40	9.0868e ⁻³	9.4985e ⁻³	9.02080e ⁴	9.42946e ⁴
$^3F_3 - ^3F_2$	7	5	E ₁	10068.77	6.4035e ⁻⁵	9.1146e ⁻⁵	0.866055e ³	1.23272e ³
$^1F_3 - ^3D_2$	7	5	E ₁	8141.46	0.010551	0.098183	9.33014e ⁴	8.68188e ⁴
$^1F_3 - ^1D_2$	7	5	E ₁	8165.07	5.2127e ⁻³	4.4878e ⁻³	4.63617e ⁴	4.33921e ⁴
$^1F_3 - ^3F_2$	7	5	E ₁	9607.45	9.8952e ⁻⁵	1.2822e ⁻⁴	1.21847e ³	1.57891e ³
$^3D_3 - ^3D_2$	7	5	E ₁	6953.22	6.5406e ⁻⁴	4.4398e ⁻⁴	4.21858e ³	2.86359e ³
$^3D_3 - ^1D_2$	7	5	E ₁	6976.83	9.6958e ⁻⁴	6.6276e ⁻⁴	6.29615e ³	4.30375e ³
$^3D_3 - ^3F_2$	7	5	E ₁	8419.20	2.4821e ⁻³	2.4699e ⁻³	2.34711e ⁴	2.33560e ⁴
$^3F_3 - ^3D_3$	7	7	E ₁	8626.11	8.4619e ⁻³	8.8455e ⁻³	5.99991e ⁴	6.27192e ⁴
$^3F_3 - ^1F_3$	7	7	E ₁	8637.51	0.135101	0.141583	9.60458e ⁶	1.00654e ⁶
$^3F_3 - ^3G_3$	7	7	E ₁	10068.26	5.1358e ⁻⁴	7.3082e ⁻⁴	4.96097e ³	7.05933e ³
$^3F_3 - ^3F_3$	7	7	E ₁	10068.45	9.7489e ⁻⁴	1.3874e ⁻³	0.941732e ⁴	1.34024e ⁴
$^1F_3 - ^3D_3$	7	7	E ₁	8164.78	0.182892	0.171161	1.16179e ⁶	1.08727e ⁶
$^1F_3 - ^1F_3$	7	7	E ₁	8176.19	7.0253e ⁻³	6.5935e ⁻³	4.47519e ⁴	4.20014e ⁴
$^1F_3 - ^3G_3$	7	7	E ₁	9606.94	1.1454e ⁻³	1.4840e ⁻³	1.00734e ⁴	1.30518e ⁴
$^1F_3 - ^3F_3$	7	7	E ₁	9607.12	2.0964e ⁻³	2.7163e ⁻³	1.84378e ⁴	2.38900e ⁴
$^3D_3 - ^3D_3$	7	7	E ₁	6976.54	0.011368	7.7688e ⁻³	5.27279e ⁴	3.60315e ⁴



Table 2 (continued)

$^3D_3 - ^1F_3$	7	7	E ₁	6987.94	5.5678e ⁻³	3.8187e ⁻³	2.59078e ⁴	1.77689e ⁴
$^3D_3 - ^3G_3$	7	7	E ₁	8418.69	0.038134	0.037943	2.57545e ⁵	2.56252e ⁵
$^3D_3 - ^3F_3$	7	7	E ₁	8418.88	0.049840	0.049591	3.36613e ⁵	3.34933e ⁵
$^3F_3 - ^1G_4$	7	9	E ₁	8613.74	0.943597	0.983425	5.18883e ⁶	5.40784e ⁶
$^3F_3 - ^3F_4$	7	9	E ₁	8637.54	0.029860	0.031297	1.65111e ⁵	1.73056e ⁵
$^3F_3 - ^3G_4$	7	9	E ₁	10068.35	0.017738	0.025243	1.33270e ⁵	1.89654e ⁵
$^1F_3 - ^1G_4$	7	9	E ₁	8152.41	0.078923	0.073635	1.33270e ⁵	1.89654e ⁵
$^1F_3 - ^3F_4$	7	9	E ₁	8176.22	0.889433	0.834711	4.40674e ⁶	4.13562e ⁶
$^1F_3 - ^3G_4$	7	9	E ₁	9607.02	0.029366	0.038049	2.00875e ⁵	2.60269e ⁵
$^3D_3 - ^1G_4$	7	9	E ₁	6964.17	9.9385e ⁻³	6.7730e ⁻³	3.57239e ⁴	2.43458e ⁴
$^3D_3 - ^3F_4$	7	9	E ₁	6987.97	0.044268	0.030350	1.60214e ⁵	1.09840e ⁵
$^3D_3 - ^3G_4$	7	9	E ₁	8418.78	1.026190	1.021040	5.39047e ⁶	5.36343e ⁶
$^3F_4 - ^3D_3$	9	7	E ₁	8798.97	7.3857e ⁻³	8.0274e ⁻³	5.44882e ⁴	5.92223e ⁴
$^3F_4 - ^1F_3$	9	7	E ₁	8810.37	4.0981e ⁻³	4.4658e ⁻³	3.03127e ⁴	3.30323e ⁴
$^3F_4 - ^3G_3$	9	7	E ₁	10241.12	3.2864e ⁻⁷	4.8389e ⁻⁷	3.28450	4.83603
$^3F_4 - ^3F_3$	9	7	E ₁	10241.31	1.3988e ⁻⁶	2.0596e ⁻⁶	13.9806	20.5853
$^3F_4 - ^1G_4$	9	9	E ₁	8786.60	0.029143	0.031587	1.66757e ⁵	1.80739e ⁵
$^3F_4 - ^3F_4$	9	7	E ₁	8810.40	0.143068	0.155902	8.23064e ⁵	8.96899e
$^3F_4 - ^3G_4$	9	9	E ₁	10241.21	1.1507e ⁻⁵	1.6943e ⁻⁵	89.4528	1.31709e ²
$^3F_4 - ^3G_5$	9	11	E ₁	8786.45	1.26585	1.37192	5.92598e ⁶	6.42251e ⁶

 $3p^5(^2P)4d-3p^5(^2P)5f$

$^3P_0 - ^3D_1$	1	3	E ₁	3274.69	0.517425	0.675747	1.23369e ⁶	1.61118e ⁶
$^3P_1 - ^3D_1$	3	3	E ₁	3183.98	0.497633	0.614294	1.12169e ⁶	1.38465e ⁶
$^3D_1 - ^3D_1$	3	3	E ₁	2212.17	0.046388	0.027674	5.04742e ⁴	3.01122e ⁴
$^1P_1 - ^3D_1$	3	3	E ₁	771.99	9.2287e ⁻³	6.6863e ⁻⁴	122.2890	88.5995
$^3P_1 - ^3D_2$	3	5	E ₁	3184.38	0.708711	0.874623	9.58722e ⁵	11.8316e ⁵
$^3P_1 - ^1D_2$	3	5	E ₁	3208.00	0.355248	0.445439	4.87721e ⁵	6.11545e ⁵
$^3P_1 - ^3F_2$	3	5	E ₁	4650.37	0.021671	0.057074	6.25213e ⁴	1.64661e ⁵
$^3D_1 - ^3D_2$	3	5	E ₁	2212.57	0.736545	0.438577	4.81024e ⁵	2.86427e ⁵

Table 2 (continued)

$^3D_1 - ^1D_2$	3	5	E ₁	2236.18	1.22293	0.744425	8.15812e ⁵	4.96601e ⁵
$^3D_1 - ^3F_2$	3	5	E ₁	3678.56	0.166502	0.274388	3.00570e ⁵	4.95328e ⁵
$^1P_1 - ^3D_2$	3	5	E ₁	772.39	0.634570	0.046056	5.05040e ⁴	3.66552e ³
$^1P_1 - ^1D_2$	3	5	E ₁	796.00	0.270633	0.020868	2.28760e ⁴	1.76394e ³
$^1P_1 - ^3F_2$	3	5	E ₁	2238.38	1.95219	1.19120	13.0485e ⁵	7.96203e ⁵
$^3P_2 - ^3D_1$	5	3	E ₁	3024.15	0.053401	0.059441	1.08587e ⁵	1.20870e ⁵
$^1D_2 - ^3D_1$	5	3	E ₁	2749.73	0.026138	0.024078	4.39417e ⁴	4.04786e ⁴
$^3F_2 - ^3D_1$	5	3	E ₁	1485.86	9.52147e ⁻⁴	2.5610e ⁻⁴	4.67391e ²	1.25718e ²
$^3D_2 - ^3D_1$	5	3	E ₁	1443.72	4.9793e ⁻⁴	1.2631e ⁻³	2.30758e ³	5.85404e ²
$^3P_2 - ^3D_2$	5	5	E ₁	3024.55	0.489832	0.545238	5.97779e ⁵	6.65395e ⁵
$^3P_2 - ^1D_2$	5	5	E ₁	3048.16	0.134021	0.151623	1.66119e ⁵	1.87938e ⁵
$^3P_2 - ^3F_2$	5	5	E ₁	4490.54	4.7222e ⁻³	1.1587e ⁻²	1.27033e ⁴	3.11705e ⁴
$^1D_2 - ^3D_2$	5	5	E ₁	2750.13	1.7185e ⁻³	1.5721e ⁻³	1.73394e ³	1.58629e ³
$^1D_2 - ^1D_2$	5	5	E ₁	2773.74	0.488528	0.457758	5.01409e ⁵	4.69828e ⁵
$^1D_2 - ^3F_2$	5	5	E ₁	4216.11	2.3801e ⁻³	5.1535e ⁻³	5.64416e ³	1.22208e ⁴
$^3F_2 - ^3D_2$	5	5	E ₁	1486.26	3.2883e ⁻⁵	8.7674e ⁻⁶	9.69027	2.58367
$^3F_2 - ^1D_2$	5	5	E ₁	1509.87	0.019205	5.3323e ⁻³	5.84077e ³	1.62170e ³
$^3F_2 - ^3F_2$	5	5	E ₁	2952.25	0.121334	0.128818	1.41078e ⁵	1.49780e ⁵
$^3D_2 - ^3D_2$	5	5	E ₁	1444.12	0.045741	0.011607	1.27258e ⁴	3.22929e ³
$^3D_2 - ^1D_2$	5	5	E ₁	1467.73	0.010317	2.7066e ⁻³	2.96522e ³	7.77843e ²
$^3D_2 - ^3F_2$	5	5	E ₁	2910.11	0.189556	0.195348	2.14155e ⁵	2.20698e ⁵
$^3P_2 - ^3D_3$	5	7	E ₁	3047.87	1.99942	2.259970	1.76987e ⁶	2.00051e ⁶
$^3P_2 - ^3F_4$	5	9	E ₁	3059.27	1.3720e ⁻⁵	1.3419e ⁻⁵	12.2360	11.9681
$^3P_2 - ^3G_3$	5	7	E ₁	4490.02	1.1601e ⁻⁷	3.3658e ⁻⁷	0.222878	0.646601
$^3P_2 - ^3F_3$	5	7	E ₁	4490.21	0.066072	0.162078	1.26941e ⁵	3.11387e ⁵
$^1D_2 - ^3D_3$	5	7	E ₁	2773.45	0.033142	0.030948	2.42925e ⁴	2.26839e ⁴
$^1D_2 - ^1F_3$	5	7	E ₁	2784.85	2.443560	2.307570	1.80580e ⁶	1.70531e ⁶
$^1D_2 - ^3G_3$	5	7	E ₁	4215.60	0.031091	0.067289	5.26512e ⁴	1.13949e ⁵
$^1D_2 - ^3F_3$	5	7	E ₁	4215.79	1.9554e ⁻⁴	4.2188e ⁻⁴	3.31169e ²	7.14482e ²

Table 2 (continued)

$^3F_2-^3D_3$	5	7	E ₁	1509.58	1.0675e ⁻³	2.9526e ⁻⁴	2.31821e ²	64.1156
$^3F_2-^1F_3$	5	7	E ₁	1520.98	0.072148	0.020323	15.9044e ³	4.48016e ³
$^3F_2-^3G_3$	5	7	E ₁	2951.73	2.688640	2.852760	2.23219e ⁶	2.36845e ⁶
$^3F_2-^3F_3$	5	7	E ₁	2951.92	0.011992	0.012686	9.95739e ³	10.5342e ³
$^3D_2-^3D_3$	5	7	E ₁	1467.44	0.156659	0.041047	3.21456e ⁴	8.42268e ³
$^3D_2-^1F_3$	5	7	E ₁	1478.84	5.1921e ⁻⁶	1.2812e ⁻⁶	1.08202	0.267006
$^3D_2-^3G_3$	5	7	E ₁	2909.59	2.6331e ⁻⁴	2.7688e ⁻⁴	2.12418e ²	2.23358e ²
$^3D_2-^3F_3$	5	7	E ₁	2909.78	2.609030	2.687620	2.10495e ⁶	2.16836e ⁶
$^3F_3-^3D_2$	7	5	E ₁	2934.16	3.4682e ⁻³	3.6346e ⁻³	3.98335e ³	4.17657e ³
$^3F_3-^1D_2$	7	5	E ₁	2957.77	0.034001	0.036226	3.69825e ⁴	4.22791e ⁴
$^3F_3-^3F_2$	7	5	E ₁	4400.14	8.8633e ⁻⁵	2.0883e ⁻⁴	2.28929e ²	5.39384e ²
$^3D_3-^3D_2$	7	5	E ₁	2696.72	0.038803	0.034332	3.76461e ⁴	3.33081e ⁴
$^3D_3-^1D_2$	7	5	E ₁	2720.33	0.018381	0.016549	1.81462e ⁴	1.63377e ⁴
$^3D_3-^3F_2$	7	5	E ₁	4162.71	5.4710e ⁻⁵	1.1534e ⁻⁴	1.26471e ²	2.66626e ²
$^1F_3-^3D_2$	7	5	E ₁	1411.28	7.5291e ⁻⁴	1.8249e ⁻⁴	2.00054e ²	48.4911
$^1F_3-^1D_2$	7	5	E ₁	1434.89	1.5911e ⁻³	3.9882e ⁻⁴	4.37054e ²	1.09546e ²
$^1F_3-^3F_2$	7	5	E ₁	2877.27	9.1313e ⁻³	9.1972e ⁻³	1.00849e ⁴	1.01576e ⁴
$^3F_3-^3D_3$	7	7	E ₁	2957.48	0.041335	0.044032	3.44518e ⁴	3.66996e ⁴
$^3F_3-^1F_3$	7	7	E₁	2968.88	0.484413	0.519994	4.06860e⁻⁵	4.36744e⁻⁵
$^3F_3-^3G_3$	7	7	E ₁	4399.63	4.9694e ⁻⁴	1.1700e ⁻³	9.16602e ²	2.15807e ³
$^3F_3-^3G_4$	7	9	E ₁	4399.82	1.19611e ⁻³	2.81727e ⁻³	2.20640e ³	5.19687e ³
$^3D_3-^3D_3$	7	7	E ₁	2720.04	0.671195	0.604166	4.73199e ⁵	4.25943e ⁵
$^1F_3-^3D_3$	7	7	E ₁	2731.45	0.021804	0.019792	1.55015e ⁴	1.40709e ⁴
$^3G_3-^3D_3$	7	7	E ₁	4162.20	4.4111e ⁻⁴	9.2973e ⁻⁴	7.28190e ²	1.53479e ³
$^3F_3-^3D_3$	7	7	E ₁	4162.38	1.2540e ⁻³	2.6432e ⁻³	2.07026e ³	4.36380e ³
$^3D_3-^1F_3$	7	7	E ₁	1434.60	0.013693	3.4295e ⁻³	2.68547e ³	6.72580e ²
$^1F_3-^1F_3$	7	7	E ₁	1446.01	0.010338	2.6324e ⁻³	2.05990e ³	5.24499e ²
$^3G_3-^1F_3$	7	7	E ₁	2876.76	0.138746	0.139698	1.09414e ⁵	1.10164e ⁵
$^3F_3-^1F_3$	7	7	E ₁	2876.94	0.183173	0.184450	1.44467e ⁵	1.45474e ⁵

Table 2 (continued)

$^1G_4-^3F_3$	9	7	E ₁	2945.11	3.27445	3.45889	2.10494e ⁶	2.22351e ⁶
$^1F_3-^3F_3$	7	7	E ₁	2968.91	0.151678	0.162826	9.90870e ⁴	10.6369e ⁴
$^3G_3-^3F_3$	7	7	E ₁	4399.72	0.019955	0.046996	2.86296e ⁴	6.74233e ⁴
$^1G_4-^3D_3$	9	7	E ₁	2707.67	0.329940	0.294298	1.79278e ⁵	1.59912e ⁵
$^1F_3-^1D_3$	7	7	E₁	2731.47	3.244960	2.945510	1.79434e⁶	1.62875e⁶
$^3G_4-^3D_3$	9	7	E ₁	4162.28	0.010225	0.021553	1.31297e ⁴	2.76741e ⁴
$^1G_4-^1F_3$	9	7	E ₁	1422.24	0.033780	8.3241e ⁻³	5.06415e ³	1.24791e ³
$^3F_4-^1F_3$	9	7	E ₁	1446.04	0.041087	0.010437	6.35682e ³	1.61759e ³
$^3G_4-^1F_3$	9	7	E ₁	2876.84	0.042757	0.043052	2.29185e ⁶	2.30768e ⁶
$^3D_3-^3F_4$	7	9	E ₁	3072.36	0.025288	0.029041	2.27462e ⁴	2.61222e ⁶
$^1F_3-^3F_4$	7	9	E ₁	3083.76	0.013998	0.016195	1.26847e ⁴	1.46758e ⁴
$^3G_3-^3F_4$	7	9	E ₁	4514.51	8.9131e ⁻⁷	2.2101e ⁻⁶	1.73101	4.29222
$^3F_3-^3F_4$	7	9	E ₁	4514.70	3.7936e ⁻⁶	9.4075e ⁻⁶	7.36825	18.2717
$^1G_4-^3F_4$	9	9	E ₁	3059.99	0.100048	0.113976	6.94306e ⁴	7.90960e ⁴
$^3F_4-^3F_4$	9	9	E ₁	3083.79	0.488672	0.565386	3.44420e ⁵	3.98488e ⁵
$^3G_4-^3F_4$	9	9	E ₁	4514.60	3.1206e ⁻⁵	7.7389e ⁻⁵	47.1440	1.16902e ²
$^3G_5-^3F_4$	11	9	E ₁	3059.84	4.34573	4.95013	2.46723e ⁶	2.81038e ⁶

 $3p^5(^2P)5d-3p^5(^2P) 5f$

$^3P_0-^3D_1$	1	3	E ₁	453.45	0.209377	0.774811	9.57195e ³	3.54216e ⁴
$^3P_1-^3D_1$	3	3	E ₁	394.05	0.231334	0.645785	7.98681e ³	2.22957e ⁴
$^3D_1-^3D_1$	3	3	E ₁	201.45	0.178299	0.131512	1.60885e ²	1.18667e ²
$^1P_1-^3D_1$	3	3	E ₁	2031.04	1.6650e ⁻⁴	7.1922e ⁻³	89.7665	3.87743e ³
$^3P_1-^3D_2$	3	5	E ₁	394.46	0.307816	0.857986	6.38941e ³	1.78094e ⁴
$^3P_1-^1D_2$	3	5	E ₁	418.07	0.168741	0.532299	3.93444e ³	1.24113e ⁴
$^3P_1-^3F_2$	3	5	E ₁	1860.44	1.1657e ⁻³	0.072606	5.38300e ²	3.35257e ⁴
$^3D_1-^3D_2$	3	5	E ₁	201.05	0.566145	0.408376	5.08813e ³	3.67021e ³
$^3D_1-^1D_2$	3	5	E ₁	177.44	0.868821	0.490659	6.0821e ³	3.43481e ³
$^3D_1-^3F_2$	3	5	E ₁	1264.94	0.011624	0.334698	2.48127e ³	7.14434e ⁴
$^1P_1-^3D_2$	3	5	E ₁	2030.63	7.6331e ⁻³	0.330500	4.11296e ³	1.78083e ⁵

Table 2 (*continued*)

$^1P_1 - ^1D_2$	3	5	E ₁	2007.02	$3.1675e^{-3}$	0.133289	$1.65540e^3$	$6.96582e^4$
$^1P_1 - ^3F_2$	3	5	E ₁	564.65	2.95240	0.437636	$5.40647e^3$	$8.01405e^2$
$^3P_2 - ^3D_1$	5	3	E ₁	300.16	0.031197	0.050370	$6.24955e^2$	$1.00904e^3$
$^1D_2 - ^3D_1$	5	3	E ₁	159.74	0.025502	0.011745	$1.44697e^2$	66.6431
$^3F_2 - ^3D_1$	5	3	E ₁	1174.16	$1.8666e^{-5}$	$4.6433e^{-4}$	3.43306	85.4003
$^3D_2 - ^3D_1$	5	3	E ₁	1187.40	$1.1134e^{-4}$	$2.8131e^{-3}$	20.9427	$5.2913e^2$
$^3P_2 - ^3D_2$	5	5	E ₁	300.56	0.286529	0.463015	$3.45312e^3$	$5.58006e^3$
$^3P_2 - ^1D_2$	5	5	E ₁	324.17	0.072547	0.137030	$1.01707e^3$	$1.92107e^3$
$^3P_2 - ^3F_2$	5	5	E ₁	1766.55	$2.0005e^{-4}$	$1.1165e^{-2}$	83.2824	$4.64855e^3$
$^1D_2 - ^3D_2$	5	5	E ₁	160.15	$1.7495e^{-3}$	$1.7031e^{-4}$	5.98575	2.63555
$^1D_2 - ^1D_2$	5	5	E ₁	183.76	0.419367	0.254938	$1.88909e^3$	$1.14840e^3$
$^1D_2 - ^3F_2$	5	5	E ₁	1626.13	$1.2372e^{-4}$	$5.8949e^{-3}$	43.6439	$2.07951e^3$
$^3F_2 - ^3D_2$	5	5	E ₁	1173.76	$2.53785e^{-7}$	$5.7118e^{-6}$	0.046644	1.04981
$^3F_2 - ^1D_2$	5	5	E ₁	1150.15	$4.1799e^{-4}$	$9.9559e^{-3}$	73.7661	$1.75696e^3$
$^3F_2 - ^3F_2$	5	5	E ₁	292.23	0.068374	0.105254	$7.7891e^2$	$1.19908e^3$
$^3D_2 - ^3D_2$	5	5	E ₁	1187.00	$1.0299e^{-3}$	0.025958	$1.93601e^2$	$4.87930e^3$
$^3D_2 - ^1D_2$	5	5	E ₁	1163.39	$2.0399e^{-4}$	$4.9660e^{-3}$	36.8332	$8.96678e^2$
$^3D_2 - ^3F_2$	5	5	E ₁	278.98	0.16037	0.161637	$1.20484e^3$	$1.67831e^3$
$^3P_2 - ^3D_3$	5	7	E ₁	323.88	1.09071	2.04629	$1.09026e^4$	$2.04545e^4$
$^3P_2 - ^1F_3$	5	7	E ₁	335.29	$2.1285e^{-5}$	$2.0972e^{-5}$	0.228017	0.224657
$^3P_2 - ^3G_3$	5	7	E ₁	1766.04	$7.2077e^{-10}$	$3.1179e^{-7}$	$2.14212e^{-4}$	$9.26655e^{-2}$
$^3P_2 - ^3F_3$	5	7	E ₁	1766.22	$2.8035e^{-3}$	0.156312	$8.33382e^2$	$4.64650e^4$
$^1D_2 - ^3D_3$	5	7	E ₁	183.47	0.029086	0.017232	93.2941	55.2712
$^1D_2 - ^1F_3$	5	7	E ₁	194.87	1.98373	1.35425	$7.17822e^3$	$4.90040e^3$
$^1D_2 - ^3G_3$	5	7	E ₁	1625.62	$1.0460e^{-3}$	0.049742	$2.63409e^2$	$1.25260e^4$
$^1D_2 - ^3F_3$	5	7	E ₁	1625.81	$1.0348e^{-5}$	$4.8135e^{-4}$	2.60647	$1.21239e^2$
$^3F_2 - ^3D_3$	5	7	E ₁	1150.44	$1.9651e^{-5}$	$4.5616e^{-4}$	3.46966	80.5411
$^3F_2 - ^1F_3$	5	7	E ₁	1139.04	$1.2192e^{-3}$	0.028436	$2.11033e^2$	$4.9218e^3$
$^3F_2 - ^3G_3$	5	7	E ₁	291.71	1.54394	2.36387	$1.25195e^4$	$1.91682e^4$

Table 2 (*continued*)

$^3F_2 - ^3F_3$	5	7	E ₁	291.90	8.8553e ⁻³	0.013310	71.8977	1.08066e ²
$^3D_2 - ^3D_3$	5	7	E ₁	1163.68	3.2024e ⁻³	0.077565	5.78529e ²	1.40124e ⁴
$^3D_2 - ^1F_3$	5	7	E ₁	1152.28	4.7716e ⁻⁷	8.5019e ⁻⁶	8.45186e ²	1.50592
$^3D_2 - ^3G_3$	5	7	E ₁	278.47	4.6617e ⁻⁴	7.0455e ⁻⁴	3.44477	5.20623
$^3D_2 - ^3F_3$	5	7	E ₁	278.66	1.579430	2.192170	1.16866e ⁴	1.62204e ⁴
$^3F_3 - ^3D_2$	7	5	E ₁	260.05	2.3790e ⁻³	2.8952e ⁻³	21.4635	26.1202
$^3F_3 - ^1D_2$	7	5	E ₁	283.66	0.020328	0.029438	2.18215e ²	3.16005e ²
$^3F_3 - ^3F_2$	7	5	E ₁	1726.04	6.1169e ⁻⁶	3.2645e ⁻⁴	2.43114	1.29746e ²
$^3D_3 - ^3D_2$	7	5	E ₁	132.28	0.044661	0.013966	1.04262e ²	32.6035
$^3D_3 - ^1D_2$	7	5	E ₁	155.90	0.017831	7.7431e ⁻³	57.8114	25.1047
$^3D_3 - ^3F_2$	7	5	E ₁	1598.27	2.8256e ⁻⁶	1.2897e ⁻⁴	96.2925	43.9535
$^1F_3 - ^3D_2$	7	5	E ₁	1220.12	1.3011e ⁻⁵	3.4697e ⁻⁴	1.84581	49.2212
$^1F_3 - ^1D_2$	7	5	E ₁	1196.51	3.5443e ⁻⁵	9.1085e ⁻⁴	4.83527	1.24260e ²
$^1F_3 - ^3F_2$	7	5	E ₁	245.86	6.1188e ⁻³	6.6093e ⁻³	49.3421	53.2980
$^3F_3 - ^3D_3$	7	7	E ₁	283.37	0.026689	0.038565	2.04222e ²	2.95096e ²
$^3F_3 - ^1F_3$	7	7	E ₁	294.78	0.276127	0.431823	2.28633e ³	3.57550e ³
$^3F_3 - ^3G_3$	7	7	E ₁	1725.53	1.7420e ⁻⁵	9.2355e ⁻⁴	4.94256	2.62029e ²
$^3D_3 - ^3D_3$	7	7	E ₁	1725.71	6.7609e ⁻⁵	3.6011e ⁻³	19.1862	1.02192e ³
$^3D_3 - ^3D_3$	7	7	E ₁	155.60	0.66331	0.28700	1.53041e ³	6.62182e ²
$^3D_3 - ^1F_3$	7	7	E ₁	167.01	1.8763e ⁻²	9.3487e ⁻³	49.8697	24.8470
$^3D_3 - ^3G_3$	7	7	E ₁	1597.76	1.3049e ⁻⁵	5.9533e ⁻⁴	3.17449	1.44820e ²
$^3D_3 - ^3F_3$	7	7	E ₁	1597.94	7.1177e ⁻⁵	3.2478e ⁻³	17.3184	7.90242e ²
$^1F_3 - ^3D_3$	7	7	E ₁	1196.80	2.7801e ⁻⁴	7.1290e ⁻³	37.9456	9.73023e ²
$^1F_3 - ^1F_3$	7	7	E ₁	1185.40	2.1711e ⁻⁴	5.4894e ⁻³	29.0712	7.35024e ²
$^1F_3 - ^3G_3$	7	7	E ₁	245.35	0.092641	0.099653	5.31394e ²	5.71616e ²
$^1F_3 - ^3F_3$	7	7	E ₁	245.53	0.122740	0.132234	7.05110e ²	7.59649e ²
$^3F_3 - ^1G_4$	7	9	E ₁	271.01	1.99015	2.630650	1.08328e ⁴	1.43192e ⁴
$^3F_3 - ^3F_4$	7	9	E ₁	294.81	0.095210	0.148895	6.13281e ²	9.59077e ²
$^3F_3 - ^1G_4$	7	9	E ₁	1725.61	9.7924e ⁻⁴	0.052094	2.16110e ²	1.14967e ⁴

Table 2 (continued)

$^3D_3 - ^1G_4$	7	9	E ₁	143.24	0.370963	0.136036	5.64078e ²	2.06854e ²
$^3D_3 - ^3F_4$	7	9	E ₁	167.04	0.2.99722	1.49444	6.19797e ³	3.09036e ³
$^3D_3 - ^3G_4$	7	9	E ₁	1597.84	2.4399e ⁻⁴	0.011132	46.1697	2.10651e ³
$^1F_3 - ^1G_4$	7	9	E ₁	1209.17	8.6121e ⁻⁴	0.022700	1.19986e ²	3.16274e ³
$^1F_3 - ^3F_4$	7	9	E ₁	1185.37	5.5263e ⁻⁴	0.013910	73.9928	1.86251e ³
$^1F_3 - ^3G_4$	7	9	E ₁	245.44	2.4960	2.68697	1.11437e ⁴	1.19960e ⁴
$^3F_4 - ^3D_3$	9	7	E ₁	353.40	0.012422	0.027723	1.47830e ²	3.29925e ²
$^3F_4 - ^1F_3$	9	7	E ₁	364.80	6.6862e ⁻³	0.015900	84.7886	2.01633e ²
$^3F_4 - ^3G_3$	9	7	E ₁	1795.55	1.2662e ⁻⁷	7.2952e ⁻⁶	3.89022e ²	2.24120
$^3F_4 - ^3F_3$	9	7	E ₁	1795.74	5.3890e ⁻⁷	3.1054e ⁻⁵	0.165593	9.54238
$^3F_4 - ^1G_4$	9	9	E ₁	341.03	0.507257	0.105420	4.37229e ²	9.08663e ²
$^3F_4 - ^3F_4$	9	9	E ₁	364.83	0.233385	0.555119	2.30226e ³	5.47604e ³
$^3F_4 - ^3G_4$	9	9	E ₁	1795.64	4.4335e ⁻⁶	2.5545e ⁻⁴	1.05946	61.0457
$^3F_4 - ^3G_5$	9	11	E ₁	340.88	2.20404	4.57671	1.55299e ⁴	3.22479e ⁴
3p⁶-3p⁵(²P) 6d								
$^1S_0 - ^3P_1$	1	3	E ₁	122767.74	6.8195e ⁻⁴	7.6847e ⁻⁴	2.28530e ⁶	2.57524e ⁶
$^1S_0 - ^3D_1$	1	3	E ₁	123145.10	0.038123	0.043045	1.28542e ⁸	1.45138e ⁸
$^1S_0 - ^1P_1$	1	3	E ₁	124500.53	0.034463	0.038929	1.18773e ⁸	1.34165e ⁸
3p⁵(²P)6d-3p⁵(²P) 4f								
$^3P_0 - ^3D_1$	1	3	E ₁	3560.20	7.7762e ⁻³	7.2481e ⁻³	6.57442e ⁴	6.12798e ⁴
$^3P_1 - ^3D_1$	3	3	E ₁	3599.09	7.4683e ⁻³	7.1141e ⁻³	2.15094e ⁴	2.04895e ⁴
$^3D_1 - ^3D_1$	3	3	E ₁	3976.44	1.9444e ⁻⁴	2.2606e ⁻⁴	6.83609e ²	7.94784e ²
$^1P_1 - ^3D_1$	3	3	E ₁	5331.87	7.5044e ⁻⁶	1.5690e ⁻⁵	47.4351	99.1782
$^3P_1 - ^3D_2$	3	5	E ₁	3598.60	9.4054e ⁻³	8.9577e ⁻³	2.70812e ⁴	2.57921e ⁴
$^3P_1 - ^1D_2$	3	5	E ₁	3553.04	6.2486e ⁻³	5.8006e ⁻³	1.75390e ⁴	1.62817e ⁴
$^3P_1 - ^3F_2$	3	5	E ₁	2113.46	8.2863e ⁻⁵	2.7219e ⁻⁵	82.2954	27.0324
$^3D_1 - ^3D_2$	3	5	E ₁	3975.95	9.2209e ⁻³	0.010721	3.24100e ⁴	3.76829e ⁴
$^3D_1 - ^1D_2$	3	5	E ₁	3930.40	0.011135	0.012650	3.82477e ⁴	4.34526e ⁴
$^3D_1 - ^3F_2$	3	5	E ₁	2490.82	8.1183e ⁻⁴	3.7039e ⁻⁴	1.11989e ³	5.10945e ²

Table 2 (continued)

$^1P_1 - ^3D_2$	3	5	E ₁	5331.38	2.6290e ⁻⁴	5.4959e ⁻⁴	1.66146e ³	3.47330e ³
$^1P_1 - ^1D_2$	3	5	E ₁	5285.82	1.4652e ⁻⁴	3.0107e ⁻⁴	9.10227e ²	1.87034e ³
$^1P_1 - ^3F_2$	3	5	E ₁	3846.24	0.021018	0.022866	6.91355e ⁴	7.52120e ⁴
$^3P_2 - ^3D_1$	5	3	E ₁	3656.30	7.5145e ⁻⁴	7.3878e ⁻⁴	1.34016e ³	1.31758e ³
$^1D_2 - ^3D_1$	5	3	E ₁	3736.47	3.1970e ⁻⁴	3.2822e ⁻⁴	5.95455e ²	6.11320e ²
$^3F_2 - ^3D_1$	5	3	E ₁	5112.67	4.4360e ⁻⁷	8.5268e ⁻⁷	1.54690	2.97342
$^3D_2 - ^3D_1$	5	3	E ₁	5117.24	2.7016e ⁻⁷	5.2026e ⁻⁶	9.43767	18.1749
$^3P_2 - ^3D_2$	5	5	E ₁	3655.82	6.8667e ⁻³	6.7496e ⁻³	1.22431e ⁴	1.20343e ⁴
$^3P_2 - ^1D_2$	5	5	E ₁	3610.26	1.9531e ⁻³	1.8721e ⁻³	3.39619e ³	3.25520e ³
$^3P_2 - ^3F_2$	5	5	E ₁	2170.68	8.8254e ⁻⁶	3.0583e ⁻⁶	5.54753	1.92243
$^1D_2 - ^3D_2$	5	5	E ₁	3735.98	2.5124e ⁻⁵	2.5807e ⁻⁵	46.7812	48.0545
$^1D_2 - ^1D_2$	5	5	E ₁	3690.42	6.1029e ⁻³	6.1122e ⁻³	1.10882e ⁴	1.11052e ⁴
$^1D_2 - ^3F_2$	5	5	E ₁	2250.84	2.4738e ⁻⁵	9.2168e ⁻⁶	16.7203	6.22937
$^3F_2 - ^3D_2$	5	5	E ₁	5112.18	9.5513e ⁻¹¹	1.8582e ⁻¹⁰	3.33003e ⁴	6.47879e ⁴
$^3F_2 - ^1D_2$	5	5	E ₁	5066.63	1.3974e ⁻⁵	2.6379e ⁻⁵	47.8559	90.4309
$^3F_2 - ^3F_2$	5	5	E ₁	3627.05	1.5587e ⁻³	1.5079e ⁻³	2.73557e ³	2.64642e ³
$^3D_2 - ^3D_2$	5	5	E ₁	5116.76	2.4947e ⁻⁵	4.8037e ⁻⁵	87.1348	1.67781e ²
$^3D_2 - ^1D_2$	5	5	E ₁	5071.20	1.7996e ⁻⁶	3.4031e ⁻⁶	6.17412	11.6755
$^3D_2 - ^3F_2$	5	5	E ₁	3631.62	2.6657e ⁻³	2.5856e ⁻³	4.69022e ³	4.54920e ³
$^3P_2 - ^3D_3$	5	7	E ₁	3610.61	0.028678	0.027496	4.98753e ⁴	4.78204e ⁴
$^3P_2 - ^1F_3$	5	7	E ₁	3588.03	4.2433e ⁻⁷	4.0655e ⁻⁷	0.728767	0.698232
$^3P_2 - ^3G_3$	5	7	E ₁	2171.77	2.1341e ⁻¹⁰	7.0993e ⁻¹¹	1.34284e ⁴	4.46701e ⁵
$^3P_2 - ^3F_3$	5	7	E ₁	2171.06	1.2354e ⁻⁴	4.2827e ⁻⁵	77.6826	2.69302
$^1D_2 - ^3D_3$	5	7	E ₁	3690.78	4.1542e ⁻⁴	0.030808	7.54925e ²	7.56552e ²
$^1D_2 - ^1F_3$	5	7	E ₁	3668.20	0.030808	0.030485	5.53021e ⁴	5.47227e ⁴
$^1D_2 - ^3G_3$	5	7	E ₁	2251.93	3.0415e ⁻⁵	1.1342e ⁻⁵	20.5770	7.67365
$^1D_2 - ^3F_3$	5	7	E ₁	2251.22	1.9165e ⁻⁶	7.1457e ⁻⁷	1.29577	0.483131
$^3F_2 - ^3D_3$	5	7	E ₁	5066.98	6.7363e ⁻⁷	1.2724e ⁻⁶	2.30725	4.35835
$^3F_2 - ^1F_3$	5	7	E ₁	5044.40	8.8379e ⁻⁶	1.6538e ⁻⁵	30.0014	56.1417

Table 2 (continued)

$^3F_2-^3G_3$	5	7	E ₁	3628.13	0.036281	0.035120	6.37121e ⁴	6.16742e ⁴
$^3F_2-^3F_3$	5	7	E ₁	3627.43	2.8543e ⁻⁴	2.7628e ⁻⁴	5.01038e ²	4.84986e ²
$^3D_2-^3D_3$	5	7	E ₁	5071.55	3.1442e ⁻⁵	5.9479e ⁻⁵	1.07889e ²	2.04090e ²
$^3D_2-^1F_3$	5	7	E ₁	5048.97	9.2389e ⁻⁹	1.7418e ⁻⁸	0.031419	0.059235
$^3D_2-^3G_3$	5	7	E ₁	3632.71	3.4592e ⁻⁵	3.3545e ⁻⁵	60.8995	59.0555
$^3D_2-^3F_3$	5	7	E ₁	3632.00	0.035423	0.034368	6.23388e ⁴	6.04810e ⁴
$^3F_3-^3D_2$	7	5	E ₁	3677.20	4.9488e ⁻⁵	4.9209e ⁻⁵	63.7644	63.4054
$^3F_3-^1D_2$	7	5	E ₁	3631.64	4.6922e ⁻⁴	4.5509e ⁻⁴	5.89696e ²	5.71935e ²
$^3F_3-^3F_2$	7	5	E ₁	2192.06	1.6520e ⁻⁶	5.8378e ⁻⁷	0.756439	0.267304
$^3D_3-^3D_2$	7	5	E ₁	3752.20	4.5261e ⁻⁴	4.6866e ⁻⁴	6.07219e ²	6.28756e ²
$^3D_3-^1D_2$	7	5	E ₁	3706.64	2.1780e ⁻⁴	2.2007e ⁻⁴	2.85146e ²	2.88118e ²
$^3D_3-^3F_2$	7	5	E ₁	2267.06	6.6947e ⁻⁷	2.5305e ⁻⁷	0.327873	0.123930
$^1F_3-^3D_2$	7	5	E ₁	5141.12	2.9669e ⁻⁷	5.7673e ⁻⁷	0.747259	1.45255
$^1F_3-^1D_2$	7	5	E ₁	5095.56	1.2521e ⁻⁶	2.3910e ⁻⁶	3.09810	5.91572
$^1F_3-^3F_2$	7	5	E ₁	3655.98	1.1960e ⁻⁴	1.1757e ⁻⁴	1.52337e ²	1.49749e ²
$^3F_3-^3D_3$	7	7	E ₁	3632.00	6.3657e ⁻⁴	6.1752e ⁻⁴	8.00178e ²	7.76233e ²
$^3F_3-^1F_3$	7	7	E ₁	3609.41	6.6031e ⁻³	6.3261e ⁻³	8.19726e ³	7.85334e ³
$^3F_3-^3G_3$	7	7	E ₁	2193.15	1.7503e ⁻⁸	6.1844e ⁻⁹	8.02221e ⁻³	6.1844e ⁻³
$^3F_3-^3F_3$	7	7	E ₁	2192.44	9.4846e ⁻⁶	3.3529e ⁻⁶	4.34431	1.53578
$^3D_3-^3D_3$	7	7	E ₁	3707.00	8.0610e ⁻³	8.1470e ⁻³	1.05555e ⁴	1.06682e ⁴
$^3D_3-^1F_3$	7	7	E ₁	3684.42	2.3404e ⁻⁴	2.3363e ⁻⁴	3.02741e ²	3.02222e ²
$^3D_3-^3G_3$	7	7	E ₁	2268.15	4.0761e ⁻⁷	1.5422e ⁻⁷	0.199818	0.075603
$^3D_3-^3F_3$	7	7	E ₁	2267.44	2.0205e ⁻⁵	7.6403e ⁻⁶	9.89898	3.74310
$^1F_3-^3D_3$	7	7	E ₁	5095.92	1.1250e ⁻⁵	2.1485e ⁻⁵	27.8383	53.1672
$^1F_3-^1F_3$	7	7	E ₁	5073.33	3.4835e ⁻⁶	6.5933e ⁻⁶	8.54377	16.1710
$^1F_3-^3G_3$	7	7	E ₁	3657.07	1.8121e ⁻³	1.7824e ⁻³	2.309e ³	2.3540e ⁻³
$^1F_3-^3F_3$	7	7	E ₁	3656.36	2.3940e ⁻³	2.3540e ⁻³	3.04989e ³	2.99882e ⁻³
$^3F_3-^1G_4$	7	9	E ₁	3655.65	0.042907	0.042167	5.46393e ⁴	5.36965e ⁴
$^3F_3-^3F_4$	7	9	E ₁	3609.37	2.4186e ⁻³	2.3171e ⁻³	3.00243e ³	2.87642e ³

Table 2 (continued)

$^3F_3-^3G_4$	7	9	E ₁	2193.04	7.0213e ⁻⁵	2.4836e ⁻⁵	32.1778	11.3823
$^3D_3-^1G_4$	7	9	E ₁	3730.65	4.3033e ⁻³	4.4049e ⁻³	5.70713e ⁻³	5.84183e ⁻³
$^3D_3-^3F_4$	7	9	E ₁	3684.38	0.039344	0.039281	5.08931e ⁴	5.08111e ⁴
$^3D_3-^3G_4$	7	9	E ₁	2268.04	1.91306e ⁻⁷	7.2376e ⁻⁸	0.093772	0.035476
$^1F_3-^1G_4$	7	9	E ₁	5119.57	2.3358e ⁻⁵	4.5019e ⁻⁵	58.3375	1.12438e ²
$^1F_3-^3F_4$	7	9	E ₁	5073.30	9.7386e ⁻⁷	1.8432e ⁻⁶	2.38849	4.52063
$^1F_3-^3G_4$	7	9	E ₁	3656.96	0.048752	0.047952	6.21278e ⁴	6.11080e ⁴
$^3F_4-^3D_3$	9	7	E ₁	3588.10	3.5497e ⁻⁴	3.3611e ⁻⁴	3.38711e ²	3.20717e ²
$^3F_4-^1F_3$	9	7	E ₁	3565.52	1.9852e ⁻⁴	1.8560e ⁻⁴	1.87047e ²	1.74879e ²
$^3F_4-^3G_3$	9	7	E ₁	2149.25	1.1898e ⁻⁷	4.0421e ⁻⁸	0.040735	0.013838
$^3F_4-^3F_3$	9	7	E ₁	2148.55	5.0014e ⁻⁷	1.6980e ⁻⁷	0.171114	0.058095
$^3F_4-^1G_4$	9	9	E ₁	3611.75	1.3839e ⁻³	1.3277e ⁻³	1.33801e ³	1.28361e ³
$^3F_4-^3F_4$	9	9	E ₁	3565.48	6.9373e ⁻³	6.4862e ⁻³	6.53622e ³	6.1112e ³
$^3F_4-^3G_4$	9	9	E ₁	2149.15	4.1772e ⁻⁶	1.4190e ⁻⁶	1.42997	0.485764
$^3F_4-^3G_5$	9	11	E ₁	3611.93	0.060385	0.057940	5.83859e ⁴	5.60218e ⁴

 $3p^5(^2P)6d-3p^5(^2P)5f$

$^3P_0-^3D_1$	1	3	E ₁	1072.69	0.130107	0.104091	9.98598e ⁴	7.98920e ⁴
$^3P_1-^3D_1$	3	3	E ₁	1111.58	0.121902	0.104739	3.34898e ⁴	2.87745e ⁴
$^3D_1-^3D_1$	3	3	E ₁	1488.93	2.1617e ⁻³	4.0299e ⁻³	1.29031e ³	1.98644e ³
$^1P_1-^3D_1$	3	3	E ₁	2844.36	7.0919e ⁻⁵	3.9930e ⁻⁵	1.27573e ²	7.18280e ²
$^3P_1-^3D_2$	3	5	E ₁	1111.18	0.154866	0.133016	4.25152e ⁴	3.65168e ⁴
$^3P_1-^1D_2$	3	5	E ₁	1087.57	0.101293	0.083278	2.66385e ⁴	2.19010e ⁴
$^3P_1-^3F_2$	3	5	E ₁	354.81	3.3567e ⁻³	2.9383e ⁻⁴	56.3749	4.93466
$^3D_1-^3D_2$	3	5	E ₁	1488.53	0.122728	0.189243	6.04616e ⁴	9.32301e ⁴
$^3D_1-^1D_2$	3	5	E ₁	1464.92	0.152964	0.228321	7.29857e ⁴	10.8942e ⁴
$^3D_1-^3F_2$	3	5	E ₁	22.55	0.395172	1.3966e ⁻⁴	44.6598	0.015783
$^1P_1-^3D_2$	3	5	E ₁	2843.96	2.4744e ⁻³	0.013926	4.44987e ³	2.50437e ⁴
$^1P_1-^1D_2$	3	5	E ₁	2820.35	1.0300e ⁻³	5.6996e ⁻³	1.82165e ³	1.00804e ⁴
$^1P_1-^3F_2$	3	5	E ₁	1377.97	0.296505	0.391452	1.25179e ⁵	1.65264e ⁵

Table 2 (*continued*)

$^3P_2 - ^3D_1$	5	3	E ₁	1168.79	0.011850	0.011261	2.15972e ³	2.05227e ³
$^1D_2 - ^3D_1$	5	3	E ₁	1248.96	4.8216e ⁻³	5.2279e ⁻³	1.00337e ³	1.08792e ³
$^3F_2 - ^3D_1$	5	3	E ₁	2625.16	4.3552e ⁻⁶	2.0863e ⁻⁵	4.00403	19.1809
$^3D_2 - ^3D_1$	5	3	E ₁	2629.73	2.6502e ⁻⁵	1.2749e ⁻⁴	24.4503	1.17617e ²
$^3P_2 - ^3D_2$	5	5	E ₁	1168.39	0.108920	0.103452	1.98362e ⁴	1.88403e ⁴
$^3P_2 - ^1D_2$	5	5	E ₁	1144.78	0.030384	0.027689	5.31215e ³	4.84103e ³
$^3P_2 - ^3F_2$	5	5	E ₁	297.59	5.2954e ⁻⁴	3.2630e ⁻⁵	6.25640	0.385516
$^1D_2 - ^3D_2$	5	5	E ₁	1248.56	3.1957e ⁻⁴	3.4815e ⁻⁴	66.4600	72.4031
$^1D_2 - ^1D_2$	5	5	E ₁	1224.95	0.092862	0.096878	1.85885e ⁴	1.93924e ⁴
$^1D_2 - ^3F_2$	5	5	E ₁	217.43	6.6147e ⁻⁴	2.1740e ⁻⁵	4.17183	0.137115
$^3F_2 - ^3D_2$	5	5	E ₁	2624.76	2.5317e ⁻¹¹	7.3057e ⁻¹¹	2.32690e ⁻⁵	6.71452e ⁻⁵
$^3F_2 - ^1D_2$	5	5	E ₁	2601.15	1.0471e ⁻⁴	4.9259e ⁻⁴	94.5186	4.44623e ²
$^3F_2 - ^3F_2$	5	5	E ₁	1158.77	0.024668	0.023026	4.41886e ³	4.12474e ³
$^3D_2 - ^3D_2$	5	5	E ₁	2629.33	2.4649e ⁻⁴	1.1856e ⁻³	2.27340e ²	1.09350e ³
$^3D_2 - ^1D_2$	5	5	E ₁	2605.72	3.6640e ⁻⁵	1.7293e ⁻⁴	33.1887	1.56677e ²
$^3D_2 - ^3F_2$	5	5	E ₁	1163.35	0.041906	0.039456	7.56601e ³	7.12365e ³
$^3P_2 - ^3D_3$	5	7	E ₁	1145.07	0.455138	0.415213	7.96125e ⁴	7.26288e ⁴
$^3P_2 - ^1F_3$	5	7	E ₁	1133.67	6.6542e ⁻⁶	6.4110e ⁻⁶	1.14089	1.109918
$^3P_2 - ^3G_3$	5	7	E ₁	297.08	8.7005e ⁻⁹	4.2716e ⁻¹⁰	7.31721e ⁻⁵	3.59244e ⁻⁶
$^3P_2 - ^3F_3$	5	7	E ₁	297.27	7.4358e ⁻³	4.5721e ⁻⁴	62.6140	3.85003
$^1D_2 - ^3D_3$	5	7	E ₁	1225.24	6.3226e ⁻³	6.6153e ⁻³	1.26622e ³	1.32485e ³
$^1D_2 - ^1F_3$	5	7	E ₁	1213.83	0.469086	0.480606	9.22024e ⁴	9.44667e ⁴
$^1D_2 - ^3G_3$	5	7	E ₁	216.92	3.0360e ⁻³	9.9327e ⁻⁵	13.6127	0.445352
$^1D_2 - ^3F_3$	5	7	E ₁	217.10	5.3399e ⁻⁵	1.7542e ⁻⁶	0.239835	7.87895e ⁻³
$^3F_2 - ^3D_3$	5	7	E ₁	2601.44	3.7739e ⁻⁶	1.7812e ⁻⁵	3.40717	16.0815
$^3F_2 - ^1F_3$	5	7	E ₁	2590.04	1.9773e ⁻⁴	9.2239e ⁻⁴	1.76957e ²	8.25470e ²
$^3F_2 - ^3G_3$	5	7	E ₁	1159.29	0.572234	0.534732	10.2595e ⁴	9.58714e ⁴
$^3F_2 - ^3F_3$	5	7	E ₁	1159.10	4.4436e ⁻³	4.1591e ⁻³	7.96448e ²	7.45449e ²
$^3D_2 - ^3D_3$	5	7	E ₁	2606.01	6.0702e ⁻⁴	2.8682e ⁻³	5.49956e ²	2.59857e ³

Table 2 (continued)

$^3D_2 - ^1F_3$	5	7	E ₁	2594.61	$1.9677e^{-7}$	$9.4221e^{-7}$	0.176718	0.846189
$^3D_2 - ^3G_3$	5	7	E ₁	1163.86	$5.2122e^{-4}$	$4.8849e^{-4}$	94.1880	88.2744
$^3D_2 - ^3F_3$	5	7	E ₁	1163.67	0.556758	0.524585	$10.0577e^4$	$9.47653e^4$
$^3F_3 - ^3D_2$	7	5	E ₁	1189.77	$7.9618e^{-4}$	$7.8364e^{-4}$	$1.07395e^2$	$1.05704e^2$
$^3F_3 - ^1D_2$	7	5	E ₁	1166.16	$7.3489e^{-3}$	$6.9488e^{-3}$	$9.52331e^2$	$9.00482e^2$
$^3F_3 - ^3F_2$	7	5	E ₁	276.21	$3.1651e^{-5}$	$1.6796e^{-6}$	0.322142	0.017095
$^3D_3 - ^3D_2$	7	5	E ₁	1264.78	$6.8187e^{-3}$	$7.5898e^{-3}$	$1.03939e^3$	$1.15692e^3$
$^3D_3 - ^1D_2$	7	5	E ₁	1241.17	$3.2323e^{-3}$	$3.4647e^{-3}$	$4.74489e^2$	$5.08605e^2$
$^3D_3 - ^3F_2$	7	5	E ₁	201.21	$1.6971e^{-5}$	$4.7810e^{-7}$	0.091664	$2.58223e^3$
$^1F_3 - ^3D_2$	7	5	E ₁	2653.69	$2.9360e^{-6}$	$1.4383e^{-5}$	1.97023	9.65154
$^1F_3 - ^1D_2$	7	5	E ₁	2630.08	$9.4769e^{-6}$	$4.5592e^{-5}$	6.24671	30.0525
$^1F_3 - ^3F_2$	7	5	E ₁	1187.71	$1.8621e^{-3}$	$1.8277e^{-3}$	$2.50303e^2$	$2.45685e^2$
$^3F_3 - ^3D_3$	7	7	E ₁	1166.45	0.01001	$9.4741e^{-3}$	$1.29839e^3$	$1.22834e^3$
$^3F_3 - ^1F_3$	7	7	E ₁	1155.05	0.104017	0.096488	$1.32236e^4$	$1.22664e^4$
$^3F_3 - ^3G_3$	7	7	E ₁	275.70	$2.8515e^{-5}$	$1.5095e^{-6}$	0.206539	0.010933
$^3F_3 - ^3F_3$	7	7	E ₁	275.89	$2.7474e^{-4}$	$1.4549e^{-5}$	1.99269	0.105522
$^3D_3 - ^3D_3$	7	7	E ₁	1241.46	0.121459	0.130254	$1.78375e^4$	$1.91292e^4$
$^3D_3 - ^1F_3$	7	7	E ₁	1230.05	$3.5333e^{-3}$	$3.7199e^{-3}$	$5.09422e^2$	$5.36324e^2$
$^3D_3 - ^3G_3$	7	7	E ₁	200.70	$3.8135e^{-5}$	$1.0688e^{-6}$	0.146374	$4.10239e^{-3}$
$^3D_3 - ^3F_3$	7	7	E ₁	200.88	$4.6772e^{-4}$	$1.3133e^{-5}$	1.79856	0.050502
$^1F_3 - ^3D_3$	7	7	E ₁	2630.37	$7.4696e^{-5}$	$3.5954e^{-4}$	49.2472	$2.37046e^2$
$^1F_3 - ^1F_3$	7	7	E ₁	2618.97	$4.7485e^{-5}$	$2.2644e^{-4}$	31.0358	$1.48005e^2$
$^1F_3 - ^3G_3$	7	7	E ₁	1188.22	0.028045	0.027551	$3.77315e^3$	$3.7067e^3$
$^1F_3 - ^3F_3$	7	7	E ₁	1188.03	0.037267	0.036600	$5.01220e^3$	$4.92249e^3$
$^3F_3 - ^1G_4$	7	9	E ₁	1178.82	0.671116	0.648423	$8.88663e^4$	$8.58614e^4$
$^3F_3 - ^3F_4$	7	9	E ₁	1155.02	0.037664	0.034937	$4.78797e^3$	$4.44135e^3$
$^3F_3 - ^3G_4$	7	9	E ₁	275.79	$3.2320e^{-3}$	$1.7105e^{-4}$	18.2188	0.964251
$^3D_3 - ^1G_4$	7	9	E ₁	1253.82	0.064143	0.070163	$9.60888e^3$	$1.05106e^4$
$^3D_3 - ^3F_4$	7	9	E ₁	1230.02	0.594499	0.625861	$8.57079e^4$	$9.02293e^4$

Table 2 (continued)

$^3D_3 - ^3G_4$	7	9	E ₁	200.78	$4.4757e^{-4}$	$1.2555e^{-5}$	1.33728	0.0375128
$^1F_3 - ^1G_4$	7	9	E ₁	2642.74	$2.2895e^{-4}$	$1.1116e^{-3}$	$1.52370e^2$	$7.39783e^2$
$^1F_3 - ^3F_4$	7	9	E ₁	2618.94	$7.3220e^{-5}$	$3.4931e^{-4}$	47.8548	$2.28306e^2$
$^1F_3 - ^3G_4$	7	9	E ₁	1188.13	0.756371	0.742961	$1.01744e^5$	$9.99405e^4$
$^3F_4 - ^3D_3$	9	7	E ₁	1122.56	$5.7234e^{-3}$	$5.0184e^{-3}$	$5.34531e^2$	$4.68692e^2$
$^3F_4 - ^1F_3$	9	7	E ₁	1111.15	$3.2126e^{-3}$	$2.7599e^{-3}$	$2.93979e^2$	$2.52553e^2$
$^3F_4 - ^3G_3$	9	7	E ₁	319.60	$1.04119e^{-6}$	$7.3998e^{-8}$	0.010133	$7.20232e^{-4}$
$^3F_4 - ^3F_3$	9	7	E ₁	319.78	$4.4290e^{-6}$	$3.1515e^{-7}$	0.043158	$3.07094e^{-3}$
$^3F_4 - ^1G_4$	9	9	E ₁	1134.93	0.022307	0.019992	$2.12955e^3$	$1.90858e^3$
$^3F_4 - ^3F_4$	9	9	E ₁	1111.12	0.112153	0.096364	$1.02621e^4$	$8.81575e^3$
$^3F_4 - ^3G_4$	9	9	E ₁	319.68	$3.6446e^{-5}$	$2.5917e^{-6}$	0.276057	0.019631
$^3F_4 - ^3G_5$	9	11	E ₁	1135.08	0.968737	0.868470	$9.25028e^4$	$8.29285e^4$

 $3p^5(^2P)7s-3p^5(^2P)7p$

$^3P_0 - ^3S_1$	1	3	E ₁	765.83	$18944e^{-3}$	$2.0730e^{-3}$	$7.41105e^2$	$8.10982e^2$
$^3P_0 - ^1P_1$	1	3	E ₁	667.65	$4.5595e^{-4}$	$3.8299e^{-3}$	$1.35569e^2$	$1.13875e^2$
$^3P_0 - ^3D_1$	1	3	E ₁	261.72	1.17438	1.19874	$1.41301e^5$	$1.44232e^5$
$^3P_0 - ^3P_1$	1	3	E ₁	742.01	1.18851	1.22319	$1.45494e^5$	$1.49739e^5$
$^1P_1 - ^1S_0$	3	1	E ₁	856.84	0.680266	0.934786	$3.33134e^5$	$4.57775e^5$
$^1P_1 - ^3P_0$	3	1	E ₁	2251.38	$3.3623e^{-4}$	$3.1707e^{-3}$	$1.13678e^3$	$1.07202e^4$
$^3P_1 - ^1S_0$	3	1	E ₁	567.83	$1.3159e^{-3}$	$8.0662e^{-4}$	94.3386	57.8274
$^3P_1 - ^3P_0$	3	1	E ₁	826.71	0.701010	0.905368	$3.19575e^5$	$4.12737e^5$
$^1P_1 - ^3S_1$	3	3	E ₁	635.58	0.426072	0.319972	$3.82684e^4$	$2.87389e^4$
$^1P_1 - ^1P_1$	3	3	E ₁	733.76	2.01484	2.03674	$2.41195e^5$	$2.43817e^5$
$^1P_1 - ^3D_1$	3	3	E ₁	1663.13	$6.13036e^{-6}$	$4.96058e^{-5}$	6.22485	50.3704
$^1P_1 - ^3P_1$	3	3	E ₁	2143.42	$1.83812e^{-6}$	$1.41187e^{-5}$	1.87762	14.4221
$^3P_1 - ^3S_1$	3	3	E ₁	789.09	$2.1882e^{-3}$	$2.5654e^{-3}$	$3.02945e^2$	$3.55171e^2$
$^3P_1 - ^1P_1$	3	3	E ₁	690.91	$2.6656e^{-4}$	$2.3672e^{-4}$	28.2922	25.1254
$^3P_1 - ^3D_1$	3	3	E ₁	238.45	1.231380	1.182360	$1.38936e^5$	$1.33405e^5$
$^3_1 - ^3P_1$	3	3	E ₁	718.75	1.20362	1.16591	$1.38249e^5$	$1.33917e^5$

Table 2 (continued)

$^1P_1 - ^1D_2$	3	5	E ₁	697.46	3.150120	2.88980	2.04429e ⁵	1.87535e
$^1P_1 - ^3P_2$	3	5	E ₁	738.67	0.963604	0.980502	7.01418e ⁴	7.13718e ⁴
$^1P_1 - ^3D_2$	3	5	E ₁	2150.73	4.2137e ⁻⁴	3.6277e ⁻³	2.60022e ²	2.23862e ³
$^3P_1 - ^1D_2$	3	5	E ₁	727.21	2.2764e ⁻⁴	2.2974e ⁻⁴	26.7667	27.0134
$^3P_1 - ^3P_2$	3	5	E ₁	686.00	2.1170e ⁻³	1.8613e ⁻³	2.21512e ²	1.94751e ²
$^3P_1 - ^3D_2$	3	5	E ₁	726.06	4.016990	3.963440	2.82497e ⁵	2.78731e ⁵
$^3P_2 - ^3S_1$	5	3	E ₁	680.34	2.164050	1.88694	2.22708e ⁵	1.94189e ⁵
$^3P_2 - ^1P_1$	5	3	E ₁	778.52	0.342431	0.394878	4.61457e ⁴	5.32134e ⁴
$^3P_2 - ^3D_1$	5	3	E ₁	1707.88	1.0526e ⁻⁶	8.9125e ⁻⁶	1.11416	9.43294
$^3P_2 - ^3P_1$	5	3	E ₁	2188.17	1.6621e ⁻³	0.0149947	1.76947e ³	1.59633e ⁴
$^3P_2 - ^1D_2$	5	5	E ₁	742.22	0.950279	0.992907	6.98379e ⁴	7.29708e ⁴
$^3P_2 - ^3P_2$	5	5	E ₁	783.43	2.81294	3.23814	2.30323e ⁵	2.61538e ⁵
$^3P_2 - ^3D_2$	5	5	E ₁	2195.48	3.2442e ⁻⁴	2.94541e ⁻³	2.08618e ²	1.89399e ³
$^3P_2 - ^3D_3$	5	7	E ₁	727.91	5.61259	5.56606	2.83375e ⁵	2.81025e ⁵
3p⁶-3p⁵(²P) 8s								
$^1S_0 - ^3P_1$	1	3	E ₁	123782.2	21.9874e ⁻³	2.1823e ⁻³	6.77069e ⁶	7.43479e ⁶
$^1S_0 - ^1P_1$	1	3	E ₁	125220.63	1.0092e ⁻³	1.1058e ⁻³	3.51846e ⁶	3.85528e ⁶
3p⁵(²P) 8s-3p⁵(²P) 7p								
$^3P_0 - ^3S_1$	1	3	E ₁	3026.76	5.1867e ⁻⁵	2.3721e ⁻⁴	3.55448e ²	1.44959e ³
$^3P_0 - ^1P_1$	1	3	E ₁	2928.58	1.2812e ⁻⁵	4.8658e ⁻⁵	73.2986	2.78365e ²
$^3P_0 - ^3D_1$	1	3	E ₁	1999.21	0.069920	0.071993	1.08508e ⁵	1.11725e ⁵
$^3P_0 - ^3P_1$	1	3	E ₁	1518.92	0.070663	0.072501	1.08744e ⁵	1.11573e ⁵
$^3P_1 - ^1S_0$	3	1	E ₁	1375.20	0.051625	0.043388	2.17078e ⁴	1.82444e ⁴
$^1P_1 - ^3P_0$	3	1	E ₁	19.34	9.3724e ⁻³	1.5588e ⁻⁶	2.33777	3.88810e ⁻⁴
$^1P_1 - ^1S_0$	3	1	E ₁	2813.61	1.3159e ⁻³	8.0662e ⁻⁴	1.14778e ²	4.01412e ²
$^1P_1 - ^3P_0$	3	1	E ₁	1419.07	0.050497	0.044960	2.26099e ⁴	2.01307e ⁴
$^3P_1 - ^3S_1$	3	3	E ₁	1596.47	0.020414	0.023207	1.15687e ⁴	1.41512e ⁴
$^3P_1 - ^1P_1$	3	3	E ₁	1498.28	0.120942	0.120453	6.03653e ⁴	6.01212e ⁴
$^3P_1 - ^3D_1$	3	3	E ₁	568.92	3.8065e ⁻⁴	1.5292e ⁻⁶	0.76400	3.06939e ⁻³

Table 2 (*continued*)

$^3P_1 - ^3P_1$	3	3	E ₁	88.63	3.2536e ⁻⁴	1.1434e ⁻⁶	0.568224	1.99687e ⁻³
$^1P_1 - ^3S_1$	3	3	E ₁	3034.87	2.1882e ⁻³	2.5654e ⁻³	1.99862e ²	8.16322e ²
$^1P_1 - ^1P_1$	3	3	E ₁	2936.69	7.4818e ⁻⁷	2.8179e ⁻⁶	1.43465	5.40337
$^1P_1 - ^3D_1$	3	3	E ₁	2007.33	0.070944	0.073704	3.70902e ⁴	3.85331e ⁴
$^1P_1 - ^3P_1$	3	3	E ₁	1527.04	0.069269	0.071712	3.59163e ⁴	3.71802e ⁴
$^3P_1 - ^1D_2$	3	5	E ₁	1534.58	0.176688	0.184169	9.25140e ⁴	9.64311e ⁴
$^3P_1 - ^3P_2$	3	5	E ₁	1493.37	0.057857	0.057454	2.86888e ⁴	2.84891e ⁴
$^3P_1 - ^3D_2$	3	5	E ₁	81.32	3.8999e ⁻⁵	1.1664e ⁻⁷	0.057336	1.71489e ⁻⁴
$^1P_1 - ^1D_2$	3	5	E ₁	2972.99	3.0693e ⁻⁶	1.2122e ⁻⁵	6.03183	23.8231
$^1P_1 - ^3P_2$	3	5	E ₁	2931.78	2.7688e ⁻⁵	1.0582e ⁻⁴	52.9161	2.02233e ²
$^1P_1 - ^3D_2$	3	5	E ₁	1519.73	0.234115	0.240293	1.20221e ⁵	1.23393e ⁵
$^3P_2 - ^3S_1$	5	3	E ₁	1580.37	0.114775	0.126952	3.82418e ⁴	4.22992e ⁴
$^3P_2 - ^1P_1$	5	3	E ₁	1482.19	0.022506	0.021782	6.59624e ³	6.38390e ³
$^3P_2 - ^3D_1$	5	3	E ₁	552.82	3.2734e ⁻⁶	9.3172e ⁻⁹	2.71978e ⁻³	7.74127e ⁻⁶
$^3P_2 - ^3P_1$	5	3	E ₁	72.53	6.1791e ⁻³	1.4396e ⁻⁵	4.33699	1.01043e ⁻²
$^3P_2 - ^1D_2$	5	5	E ₁	1518.49	0.057838	0.058849	1.77914e ⁴	1.81024e ⁴
$^3P_2 - ^3P_2$	5	5	E ₁	1477.28	0.182521	0.176820	5.31382e ⁴	5.14786e ⁴
$^3P_2 - ^3D_2$	5	5	E ₁	65.22	1.3450e ⁻³	2.5343e ⁻⁶	0.763354	1.43832e ⁻³
$^3P_2 - ^3D_3$	5	7	E ₁	1532.80	0.325048	0.339390	1.01880e ⁵	1.06376e ⁵

Table 3: Comparison between experimental and theoretical values of the oscillator strength for some transitions from the ground state $3p^6\ ^1S_0$ to $3p^5\ n'l'$ transition arrays.

Transition / Reference	Experimental	Theoretical		
	f_{ij}	f_{ij}	$f_{ij}(\nu)$	$f_{ij}(l)$
$3p^6\ ^1S_0 - 3p^57s\ ^3P_1$				
Present Work			0.0056279	0.0061725
Zatsarinny <i>et.al.</i> [39]			0.0058	0.0068
Lee <i>et.al.</i> [76]	0.037 ± 0.01	$\begin{cases} 0.013 \\ 0.03 \end{cases}$		
Wu <i>et.al.</i> [33]	0.0139			
$3p^6\ ^1S_0 - 3p^57s\ ^1P_1$				
Present Work			0.003071	0.0033239
Lee <i>et.al.</i> [76]		0.00074		
$3p^6\ ^1S_0 - 3p^55d\ ^3P_1$				
Present Work			0.0010364	0.0011706
Zatsarinny <i>et.al.</i> [39]			0.0025	0.0022
Lee <i>et.al.</i> [76]	0.0032 ± 0.0003	0.0043		
Wu <i>et.al.</i> [33]	0.0043			
Chan <i>et. al.</i> [30]	0.0041			
$3p^6\ ^1S_0 - 3p^55d\ ^3D_1$				
Present Work			0.053039	0.060041
Zatsarinny <i>et.al.</i> [39]			0.0317	0.0278
Lee <i>et.al.</i> [76]	0.037 ± 0.01	0.03		
Wu <i>et.al.</i> [33]	0.0043			
Chan <i>et. al.</i> [30]	0.0041			
$3p^6\ ^1S_0 - 3p^55d\ ^1P_1$				
Present Work			0.068294	0.077339
Wu <i>et.al.</i> [33]	0.0411			
Chan <i>et. al.</i> [30]	0.0426			

$3p^6\ ^1S_0 - 3p^56d\ ^3D_1$

Present Work

0.038123

0.043045

Lee *et.al.* [76]

0.029

Table 4: Comparison between experimental and theoretical values of the life times (ns) of some levels in neutral argon atom

Level / Reference	Theoretical values				Experimental values
	T_v	T_l	$\tau_{radiative}$	$\tau_{Multiconfiguration}$	$\tau_{radiative}$
5d[1/2]₁					
This work	295.10451	261.12582			
Das <i>et. al.</i> [41]					73.2 ± 4.7
Zurro <i>et. al.</i> [78]			90		77 ± 7
Gruzdev <i>et. al.</i> [79]			231	111	
5d' [3/2]₁					
This work	4.92484	3.83055			
Das <i>et. al.</i> [41]					87.8 ± 5.1
Gruzdev <i>et. al.</i> [79]			3.48	0.97	

Table 5: Relativistic Landé g -factor and hyperfine structure constants of levels belonging to the intermediate Rydberg series series $nf[K]_J$ ($n=4, 5$), $nd[K]_J$ ($n=5, 6$), $np[K]_J$ ($n=6, 7$) and $ns[K]_J$ ($n=7, 8$) relative to the ground state $3p^6\ ^1S_0$ for neutral argon atom spectra .

Designation	LS-Term	g_J	A_{MD} [MHz]	B_{QE} [MHz]
4f				
$4f\ [3/2]_1$	3D_1	0.49897	-178.17379	11.65133
$4f\ [3/2]_2$	3D_2	1.10135	-106.20611	22.71935
$4f\ [5/2]_2$	$^1D_2^\dagger$	0.85455	4.491163	-69.12849
$4f\ [5/2]_3$	$^3D_3^\dagger$	1.18328	3.69764	-87.1126
$4f\ [7/2]_3$	1F_3	0.95937	64.76491	-30.04807
$4f\ [7/2]_4$	3F_4	1.02225	87.17149	108.87556
$4f\ [9/2]_4$	1G_4	1.19110	50.33734	-33.79020
$4f\ [9/2]_5$	3G_5	1.20048	71.29366	116.52313
$4f'\ [5/2]_2$	$^3F_2^\dagger$	0.87713	-201.19923	-3.543188
$4f'\ [5/2]_3$	$^3F_3^\dagger$	0.82572	195.68097	-4.64485
$4f'\ [7/2]_3$	3G_3	1.19859	-143.78531	-4.45522
$4f'\ [7/2]_4$	3G_4	1.08719	152.13419	-5.18552
5d				
$5d\ [1/2]_1$	3P_1	1.38562	33.07150	-28.40964
$5d\ [7/2]_4$	3F_4	1.25051	89.11551	116.38237
$5d\ [7/2]_3$	$^3F_3^\dagger$	1.06789	119.13155	116.14584
$5d\ [3/2]_2$	$^3P_2^\dagger$	1.34760	21.87261	-57.95466
$5d\ [3/2]_1$	3D_1	0.75270	-95.55501	-8.35450
$5d\ [5/2]_2$	3F_2	0.95344	109.37551	1.031167
$5d\ [5/2]_3$	$^1F_3^\dagger$	1.22939	71.37795	-20.19362
$5d'\ [5/2]_2$	1D_2	0.780744	275.79377	-14.36251
$5d'\ [5/2]_3$	3D_3	1.120168	201.61418	-8.76123
$5d'\ [3/2]_2$	$^3D_2^\dagger$	1.25218	-166.3056	-12.00377
$5d'\ [3/2]_1$	1P_1	0.861759	-301.93397	1.80797

Table 5 (*continued*)**6p**

$6p[1/2]_1$	3S_1	1.83764	184.54045	1.64734
$6p[5/2]_3$	3D_3	1.33401	121.08902	114.00754
$6p[5/2]_2$	3D_2	1.10797	180.21100	96.81019
$6p[3/2]_1$	3D_1	1.03543	218.62299	13.59850
$6p[3/2]_2$	3P_2	1.30928	151.66750	60.01242
$6p'[3/2]_1$	1P_1	0.61817	566.62391	-2.56897
$6p'[3/2]_2$	1D_2	1.25066	282.65472	-42.81507
$6p'[1/2]_1$	3P_1	1.51068	-245.77653	-1.06435

7s

$7s[1/2]_1$	1P_1	1.17753	291.58747	54.46992
$7s[3/2]_1$	3P_1	1.32329	620.17655	3.77657
$7s[3/2]_2$	3P_2	1.50102	181.27813	116.49156

5f

$5f[3/2]_1$	3D_1	0.49897	-178.19278	11.65215
$5f[3/2]_2$	3D_2	1.10188	-105.78867	22.36134
$5f[5/2]_2$	$^1D_2^\dagger$	0.84836	5.44505	-70.51700
$5f[5/2]_3$	$^3D_3^\dagger$	1.17923	4.68600	-89.30427
$5f[7/2]_3$	1F_3	0.96174	64.20318	-32.33295
$5f[7/2]_4$	3F_4	1.02228	87.18350	108.95269
$5f[9/2]_4$	1G_4	1.19291	49.88223	-36.42173
$5f[9/2]_5$	3G_5	1.20040	71.28950	116.52661
$5f'[5/2]_2$	$^3F_2^\dagger$	0.88279	-202.60180	-1.79142
$5f'[5/2]_3$	$^3F_3^\dagger$	0.82338	196.16102	-2.35422
$5f'[7/2]_3$	3G_3	1.20262	-144.72142	-2.25974
$5f'[7/2]_4$	3G_4	1.08536	152.56035	-2.62219

6d

$6d[1/2]_1$	3P_1	1.22043	44.36623	-31.02924
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Table 5 (*continued*)

$6d [7/2]_4$	3F_4	1.25051	89.11173	116.4388
$6d [7/2]_3$	$^3F_3^\dagger$	1.06885	118.95235	116.41058
$6d [3/2]_2$	$^3P_2^\dagger$	1.32894	27.95451	-62.99269
$6d [3/2]_1$	3D_1	0.79509	-104.82013	-4.65789
$6d [5/2]_2$	3F_2	0.96411	106.54658	-4.97696
$6d [5/2]_3$	$^1F_3^\dagger$	1.23241	70.41869	-24.19346
$6d' [5/2]_2$	1D_2	0.77113	27.74141	-8.32763
$6d' [5/2]_3$	3D_3	1.11618	202.73747	-4.94686
$6d' [3/2]_2$	$^3D_2^\dagger$	1.26979	-169.39135	-6.97151
$6d' [3/2]_1$	1P_1	0.84502	-304.3269	7.29958
7p				
$7p[1/2]_1$	3S_1	1.81382	192.94165	1.79005
$7p[5/2]_3$	3D_3	1.33401	121.08902	114.00754
$7p[5/2]_2$	3D_2	1.09416	174.32686	107.40585
$7p[3/2]_1$	3D_1	1.04324	216.52217	11.70944
$7p[3/2]_2$	3P_2	1.40417	124.02187	0.10882
$7p' [3/2]_1$	1P_1	0.61703	586.62472	-0.99346
$7p' [3/2]_2$	1D_2	1.16958	305.6366	-1.97707
$7p' [1/2]_1$	3P_1	1.52785	-273.09577	-0.87494
8s				
$8s[1/2]_1$	1P_1	1.17058	295.04723	56.93358
$8s[3/2]_1$	3P_1	1.33025	614.90518	1.32465
$8s[3/2]_2$	3P_2	1.50102	179.31350	116.51648

Table 6: Comparison of Relativistic Landé g -factor of $4f [5/2]_3 {}^3D_3$, $6p [5/2]_2 {}^3D_2$, $6p [3/2]_1 {}^3D_1$, $6d [1/2]_1 {}^3P_1$ and $6d [3/2]_1 {}^3D_1$ with other experimental and NIST results.

Designation	This work	NIST [48]	Experimental work [15]
$4f [5/2]_3 {}^3D_3$	1.18328	1.184	
$6p [5/2]_2 {}^3D_2$	1.10797	1.18	
$6p [3/2]_1 {}^3D_1$	1.03543	1.067	
$6d [1/2]_1 {}^3P_1$	1.21798		1.22
$6d [3/2]_1 {}^3D_1$	0.79509		0.91